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本論文係洪睿帆君（R13942124）在國立臺灣大學電信工程學研究所完成之碩士學位論文，於民國 109 年 5 月 1 日承下列考試委員審查通過及口試及格，特此證明

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首先感謝我的爸媽，

摘要

中文摘要

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Abstract

Abstract

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Chapter 1 Introduction

As the demand of high speed and low latency communication networks, future wireless systems must support higher spectral efficiency, flexible resource sharing, and massive connectivity. To increase the channel capacity and reliability, multiple access(MA) techniques become more important. It can be classified into two categories: Orthogonal multiple access (OMA) and non-orthogonal multiple access (NOMA). There are many OMA techniques, such as time division multiple access in two generation (2G), Code Division Multiple Access (CDMA) in third generation (3G) and Orthogonal Frequency Division Multiple Access (OFDMA) in fourth generation (4G). [1] However, Orthogonal multiple access (OMA) avoids inter-user interference by assigning disjoint resources, its user capacity is fundamentally limited by the number of orthogonal resource units [2].

In particular, beyond 5G and toward 6G, Non-Orthogonal multiple access (NOMA) is regarded as a potential multiple access technique to support the growing demands for massive connectivity and diverse service requirements. [3]. In NOMA systems, multiple users are allowed to transmit over the same resource, relaxing the one-user-per-resource constraint and improving resource utilization at the cost of receiver-side interference management. Different NOMA schemes rely on different user-separation principles. For example, gain division multiple access (GDMA) separates users by exploiting their complex channel gains and is typically associated with joint MAP or ML detection

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[4]. Power-domain NOMA (PD-NOMA) mainly relies on received-power disparity and usually adopts successive interference cancellation (SIC) [2, 5]. Code-domain schemes such as SCMA, low density signature CDMA (LDS-CDMA) and HDS-CDMA exploit sparse or structured signatures for user separation[6, 7]. Pattern division multiple access (PDMA) maps users onto shared orthogonal resources according to a predefined pattern matrix [8].

In uplink NOMA, the base station (BS) receives the superimposed signals from multiple users and performs multiuser detection (MUD) to separate the users. That is, the receiver structure is often the key factor that determines both reliability and practicality.

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Chapter 2 Single resource systems

This chapter studies uplink NOMA with single resource system, where PD-NOMA with soft SIC and GDMA [4] are discussed. We assume that the channel state information (CSI) is perfectly known at the receiver. The performance of these two detection schemes is evaluated in terms of bit error rate (BER) under different large scale fading coefficients.

2.1 Single resource system model

Consider an uplink system where perfect timing synchronization is assumed. U users transmit simultaneously over one shared resource, which may be a frequency band, time slot, or spreading signature. The received signal at the BS is

$$y = \sum_{k=1}^{U} \sqrt{P_k \beta_k} h_k x_k + n, \tag{2.1}$$

where P_k is the transmit power of user k, β_k is the large-scale fading coefficient of user k, h_k is the small-scale Rayleigh flat-fading coefficient of user k, x_k is transmission symbol of user k and $n \sim \mathcal{CN}(0, N_0)$ is complex AWGN. Equivalently, the noise variance per dimension is $\sigma^2 = N_0/2$. In this work, for simplicity, $P_k = 1$ is assumed unless otherwise stated.

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2.2 Detection principle for PD NOMA : soft SIC

In PD-NOMA, users are mainly separated by their received-power difference. The instantaneous received-power metric of user k can be written as

$$\rho_k = P_k \beta_k |h_k|^2 \triangleq |g_k|^2. \tag{2.2}$$

The SIC detection order is denoted by

$$\pi = \{\pi(1), \pi(2), \ldots, \pi(U)\}, \tag{2.3}$$

where $\pi(i)$ is the user detected at stage i. For received power-based ordering, this order is usually determined such that $\rho_{\pi(1)} \geq \rho_{\pi(2)} \geq \cdots \geq \rho_{\pi(U)}$. The user indices are then relabeled according to this order. For simplicity, we assume the user indices in (2.1) have been renumbered, so that the i^{th} SIC stage detects the LLR of user i, denoted L_{x_i}.

At the first SIC stage, the LLR of x_1, L_{x_1}, is obtained by regarding signals from other users as approximately Gaussian in (2.1) and the soft estimate is denoted as $\hat{s}_1$. At the $(i+1)^{th}$ SIC stage, the LLR of x_{i+1}, $L_{x_{i+1}}$, is obtained by a residual signal $y_{(i+1)}$ which is obtained by subtracting the contribution of the soft estimate $\hat{s}_i$ in the i^{th} stage and regarding signals from other users as Gaussian. Therefore,

$$y_{(1)} = y, \tag{2.4}$$

$$y_{(i+1)} = y_{(i)} - g_i \hat{s}_i. \tag{2.5}$$

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For BPSK modulation, the log likelihood ratio (LLR) of user i can be simplified as

$$L_{x_i} = \log \frac{p(y_{(i)}|x_{\pi(i)} = +1)}{p(y_{(i)}|x_{\pi(i)} = -1)} = \frac{4\,\mathrm{Re}\{g_i^* y_{(i)}\}}{\sigma^2_{\mathrm{eff},(i)}}. \tag{2.6}$$

The soft estimate of the detected symbol given $y_{(i)}$ is expressed [9] by

$$\hat{s}_i = \mathbb{E}\left[x_i|L_{x_i}\right] = \tanh\left(\frac{L_{x_i}}{2}\right). \tag{2.7}$$

and the conditional variance is obtained [9] as

$$\sigma^2_{\hat{s}_i} = \mathbb{E}\left[x_i^2|L_{x_i}\right] = 1 - \tanh^2\left(\frac{L_{x_i}}{2}\right). \tag{2.8}$$

Therefore, the effective noise variance is

$$\sigma^2_{\mathrm{eff},(i)} = 2\sigma^2 + \sum_{j=i+1}^{U} |g_j|^2 + \sum_{j=1}^{i-1} \sigma^2_{\hat{s}_j}. \tag{2.9}$$

The first term is the noise variance, the second term is the interference from undetected users, and the third term is the uncertain residual interference from previously detected users. This soft cancellationprocedure is repeated until all users are detected.

2.3 Detection principle for GDMA

In GDMA, users are distinguished by their distinct channel gains. Unlike PD-NOMA, which relies mainly on received-power differences, GDMA uses both the magnitude and phase of g_k.

At the receiver, the superimposed signal is first used to calculate the *a posteriori* probabilities (APPs) of all possible combinations of multiuser symbols, and then the LLR

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Table 2.1: Mapping rules for superimposed signal in the case of $M = 1$ and $U = 2$

l	x_1	x_2	s_l
0	1	1	$\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$
1	1	-1	$\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$
2	-1	1	$-\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$
3	-1	-1	$-\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$

of each user is then obtained accordingly [4]. There are 2^{MU} possible superimposed signal levels, denoted by the set

$$\mathcal{S} = \{s_0, s_1, \ldots, s_{2^{MU}-1}\}, \tag{2.10}$$

where M is the number of coded bits per modulation symbol for each user and $s_l \in \mathcal{S}$ contributes to the signal part in (2.1). Take BPSK modulation ($M = 1$) as an example, the superimposed signal

$$\begin{aligned} s_l &= (-1)^{l_2[1]} \cdot \sqrt{\beta_1}h_1 + (-1)^{l_2[2]} \cdot \sqrt{\beta_2}h_2 + \cdots + (-1)^{l_2[U]} \cdot \sqrt{\beta_U}h_U \\ &= \sum_{k=1}^{U} (-1)^{l_2[k]} \cdot \sqrt{\beta_k}h_k, \quad l = 0, 1, \ldots, 2^U - 1, \end{aligned} \tag{2.11}$$

where $l_2[k]$ denotes the transmission bit of user k in the binary representation.

In detail:

$$\begin{aligned} l = 0 &\rightarrow (l_2[1], l_2[2], \ldots, l_2[U]) = (0, 0, \ldots, 0) \\ l = 1 &\rightarrow (l_2[1], l_2[2], \ldots, l_2[U]) = (0, 0, \ldots, 1) \\ &\vdots \\ l = 2^U - 1 &\rightarrow (l_2[1], l_2[2], \ldots, l_2[U]) = (1, 1, \ldots, 1). \end{aligned} \tag{2.12}$$

The superimposed signal levels s_l for the case $U = 2$ are listed in Table 2.1, where $l \in \{0, 1, 2, 3\}$.

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The receiver computes the APPs of each possible superimposed signal $s_l \in \mathcal{S}$ by

$$P(s_l|y) = P(y|s_l) \cdot \frac{P(s = s_l)}{P(y)}. \tag{2.13}$$

Assuming the signals in $\mathcal{S}$ are equally probable, the APPs can be formulated as

$$P(s_l|y) \propto \exp\left(-\frac{|y - s_l|^2}{2\sigma^2}\right). \tag{2.14}$$

The log-likelihood ratio (LLR) of user k is then calculated by

$$L_{x_k} = \log \frac{\sum\limits_{x_k=+1} \exp\left(-\frac{|y-s_l|^2}{2\sigma^2}\right)}{\sum\limits_{x_k=-1} \exp\left(-\frac{|y-s_l|^2}{2\sigma^2}\right)}. \tag{2.15}$$

After the LLRs of all users derived, we then use hard decision to obtain the detected symbols of all users.

2.4 Simulations of PD NOMA with soft SIC and GDMA

The single-resource simulations use BPSK modulation for each user over independent Rayleigh flat-fading channels. Figures 2.1, 2.2, and 2.3 show the BER performance of two users with different large-scale fading coefficients (β_1, β_2). As the large-scale fading gap becomes larger, the soft SIC detection converges toward the performance of the GDMA detection, indicating that the received-power disparity provides a clearer detection order and reduces the ambiguity of early cancellation stages. This observation also suggests that the SIC gain mainly stems from the received-power disparity, which is determined by both the transmit power P_k and the large-scale fading coefficient β_k. In practice, power control can compensate for large-scale fading differences; however, excessive com-

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pensation weakens the power-domain separation required by soft SIC, and this benefit is achieved at the severe cost of user fairness [10].

Figure 2.4 shows the BER performance of three users with large-scale fading coefficients $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$. This result further clarifies the limitation of soft SIC detection. As the number of users increases, soft SIC becomes more susceptible to error propagation because each cancellation stage depends on the reliability of the previous soft estimates. In contrast, GDMA evaluates the superimposed signals and can therefore maintain favorable BER performance even as the number of users increases [11, 12]. This performance advantage comes with an exponential growth cost with U.

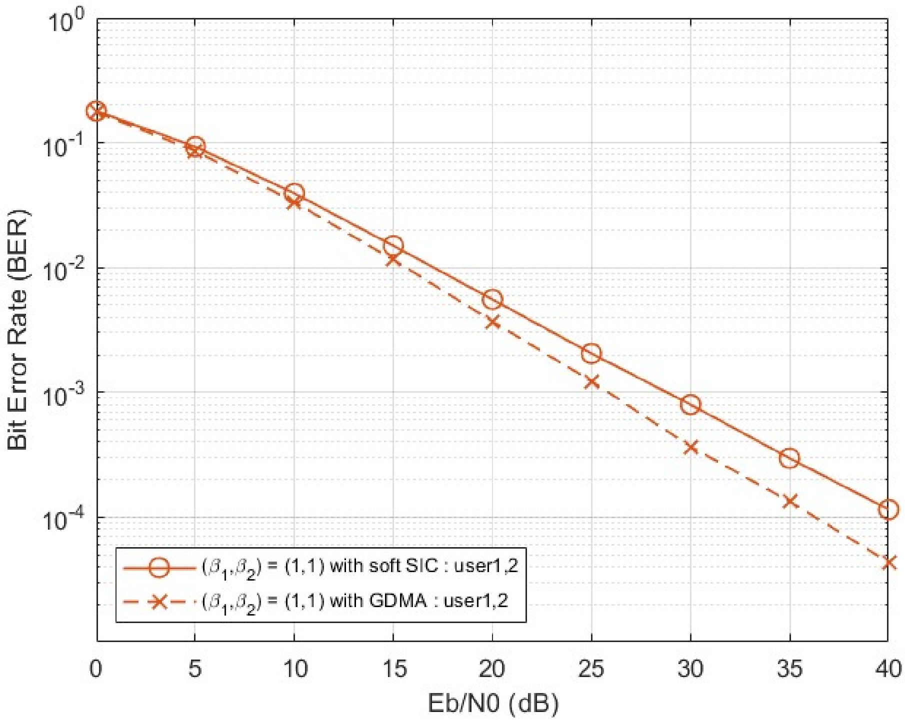

Figure 2.1: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 1)$.

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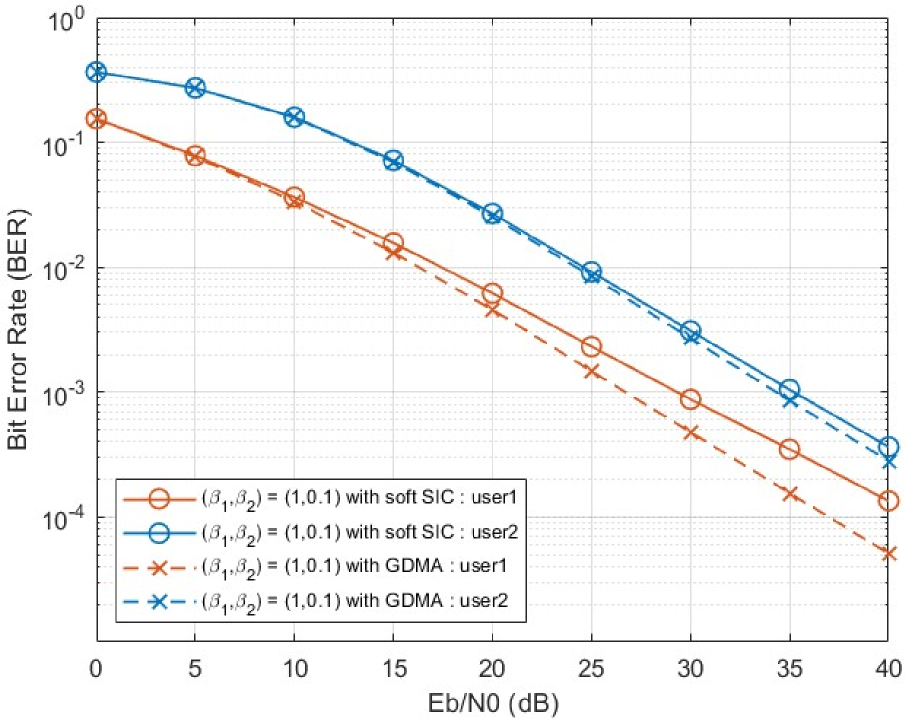

Figure 2.2: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 0.1)$.

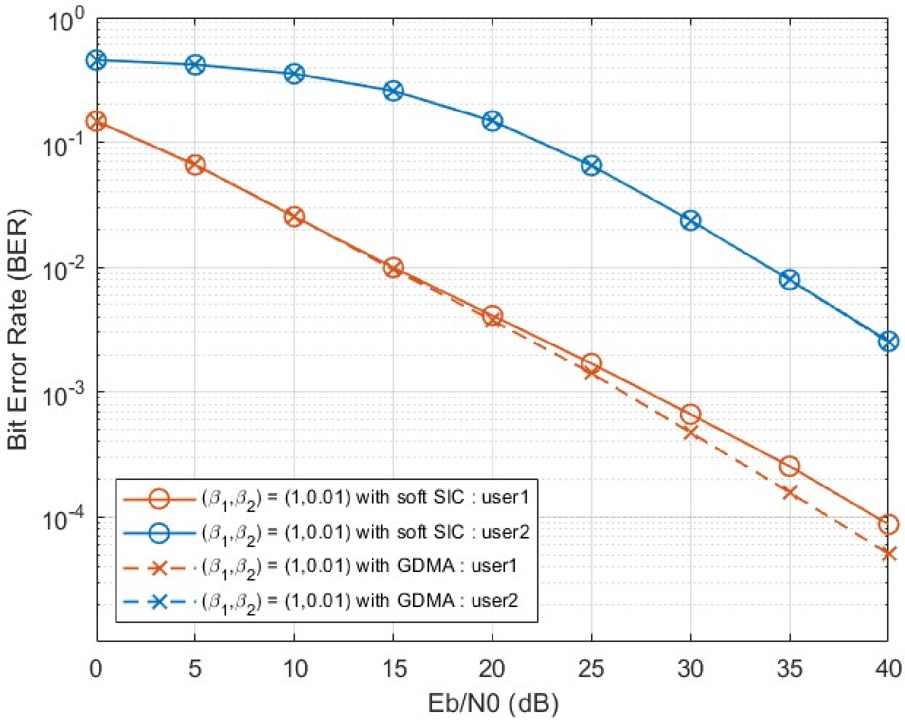

Figure 2.3: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 0.01)$.

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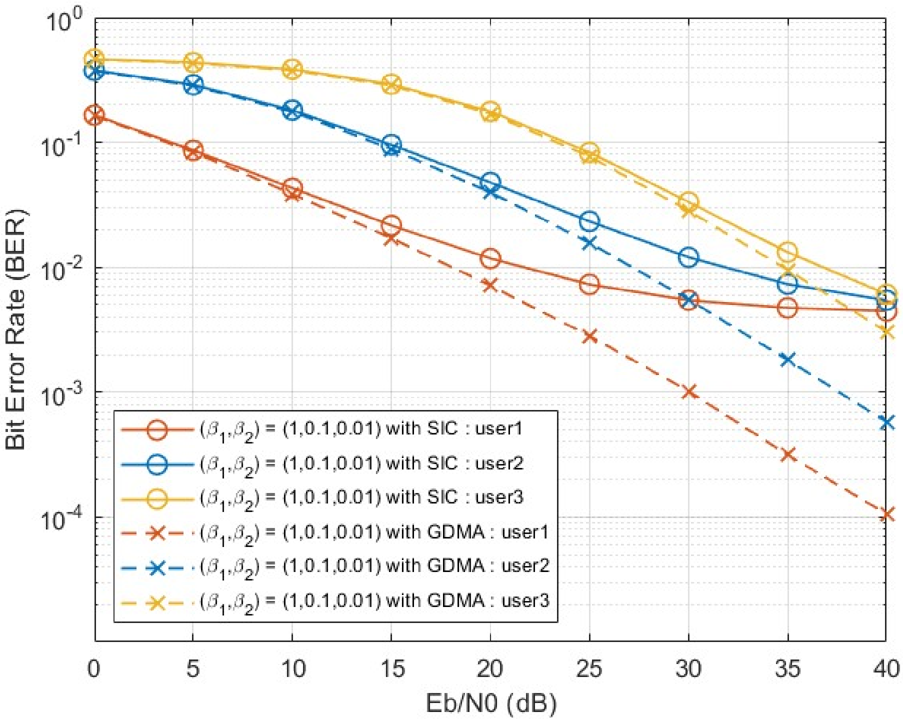

Figure 2.4: Users' BERs for single resource systems of $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

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Chapter 3 Multiple resource NOMA systems

In this chapter, we extend single-resource PD-NOMA to a multiple-resource system, where several shared-resources are used to improve multiuser separation ability. Since single-resource PD-NOMA mainly relies on received-power differences, the multiple-resource model allows the BS to further exploit resource domain channel-vector information. For multi-resource PD-NOMA, MMSE-SIC detection is adopted, post-SINR ordering and amplitude-based ordering are considered as two detection-ordering methods, and the BER performance is compared with HDS-CDMA with ML detection [7].

3.1 Multi-resource PD-NOMA system model

Multi-resource PD-NOMA refers to an architecture in which each user can transmit over multiple shared resources. At one symbol time, the received vector can be written as

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n}, \tag{3.1}$$

where $\boldsymbol{y} \in \mathbb{C}^{R\times 1}$ are the received signals over R resources, $\boldsymbol{x} \in \mathbb{C}^{U\times 1}$ are the transmitted signals of U users and $\boldsymbol{n} \in \mathbb{C}^{R\times 1}$ denotes complex AWGN. The channel matrix $\boldsymbol{H} \in$

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$\mathbb{C}^{R\times U}$ is

$$\boldsymbol{H} = \begin{bmatrix} \sqrt{P_1\beta_1}h_{1,1} & \sqrt{P_2\beta_2}h_{1,2} & \cdots & \sqrt{P_U\beta_U}h_{1,U} \\ \sqrt{P_1\beta_1}h_{2,1} & \sqrt{P_2\beta_2}h_{2,2} & \cdots & \sqrt{P_U\beta_U}h_{2,U} \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{P_1\beta_1}h_{R,1} & \sqrt{P_2\beta_2}h_{R,2} & \cdots & \sqrt{P_U\beta_U}h_{R,U} \end{bmatrix}. \tag{3.2}$$

The kth column $\boldsymbol{h}_k$ of $\boldsymbol{H}$ represents the resource domain channel vector of user k. For the representative two-resource and three-user case, the received signal is

$$\begin{bmatrix} y_A \\ y_B \end{bmatrix} = \begin{bmatrix} h_{A,1} & h_{A,2} & h_{A,3} \\ h_{B,1} & h_{B,2} & h_{B,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} n_A \\ n_B \end{bmatrix}, \tag{3.3}$$

where the large-scale fading and transmit-power factors are absorbed into each $h_{r,k}$.

We employ MMSE-SIC detection for (3.1). The resulting filter structure is similar to those widely used in multi-antenna systems [13, 14] and pattern division multiple access (PDMA) receivers [8]. In multi-antenna reception, user separation mainly comes from spatial observations, while in PDMA it is further assisted by user-specific pattern mapping across resources. In contrast, the considered multi-resource PD-NOMA system does not adopt sparse signatures or pattern design [15]; instead, it relies on the channel vectors across shared resources to improve user separation reliability and overall performance.

3.2 MMSE-SIC Detection

The MMSE-SIC receiver provides a low-complexity alternative to joint ML/MAP detection [16] by applying a linear MMSE filter and successively cancelling users. The main quantities needed after filtering are the desired effective channel gain, the residual

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interference-plus-noise variance, the LLR, and the soft estimate used for cancellation. For the multi-resource model of (3.1), consider user k be detected. The MMSE filter is designed by following [8, Eqs. (6), (7)] with the coefficient

$$\boldsymbol{w}_k^H = \boldsymbol{h}_k^H \left(\boldsymbol{H}\boldsymbol{H}^H + 2\sigma^2\boldsymbol{I}\right)^{-1}, \tag{3.4}$$

The filter output of user k can be formulated as

$$z_k = \boldsymbol{w}_k^H\boldsymbol{y} = \boldsymbol{w}_k^H\boldsymbol{h}_k x_k + \sum_{j\neq k}\boldsymbol{w}_k^H\boldsymbol{h}_j x_j + \boldsymbol{w}_k^H\boldsymbol{n}. \tag{3.5}$$

In (3.5), we define the effective channel gain:

$$a_k = \boldsymbol{w}_k^H\boldsymbol{h}_k, \tag{3.6}$$

and the effective residual interference-plus-noise variance:

$$\sigma_{k,\text{eff}}^2 = \sum_{j\neq k}|\boldsymbol{w}_k^H\boldsymbol{h}_j|^2 + 2\sigma^2\|\boldsymbol{w}_k^H\|^2. \tag{3.7}$$

3.2.1 Detection Ordering

To determine successive detection orders in the SIC process, two ordering strategies are considered. The first is post-SINR ordering [14]. The post-filtering SINR of user k is

$$\gamma_k^{\text{post}} = \frac{\left|\boldsymbol{w}_k^H\boldsymbol{h}_k\right|^2}{\sum_{j\neq k}\left|\boldsymbol{w}_k^H\boldsymbol{h}_j\right|^2 + 2\sigma^2\left\|\boldsymbol{w}_k\right\|^2}. \tag{3.8}$$

The receiver detects the user with the largest post-SINR first. This criterion uses both the magnitude and phase of the channel information, and it reflects the actual quality of each user after MMSE filtering, which are generally preferred in reducing SIC error propaga-

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tion.

The second strategy is amplitude-based ordering. Users are sorted according to a channel magnitude or received-power metric such as

$$\gamma_k^{\text{amp}} = \|\boldsymbol{h}_k\|^2. \tag{3.9}$$

This ordering is simpler because it relies only on channel magnitudes, but ignores how MMSE filtering suppresses multiuser interference. After determining the SIC orders and then re-numbering the user indexes, we assume

$$\gamma_1 \geq \ldots \geq \gamma_U \tag{3.10}$$

for simplicity. This renumbering is applied for either one of the ordering strategies and simplifies the superscript.

3.2.2 SIC Procedure

After the detection order is decided, users are renumbered as $1, 2, \ldots, U$. At stage i, the residual vector is $\boldsymbol{y}_{(i)}$ and the corresponding channel matrix is

$$\boldsymbol{H}_{(i)} = [\boldsymbol{h}_i, \boldsymbol{h}_{i+1}, \ldots, \boldsymbol{h}_U]. \tag{3.11}$$

The MMSE filter output and the effective variance of the residual interferences and noise of user i can be derived as

$$z_i = \boldsymbol{w}_i^H \boldsymbol{y}_{(i)} = \boldsymbol{w}_i^H \boldsymbol{h}_i x_i + \sum_{j>i} \boldsymbol{w}_i^H \boldsymbol{h}_j x_j + \boldsymbol{w}_i^H \boldsymbol{n}. \tag{3.12}$$

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$$\sigma_{i,\text{eff}}^2 = \sum_{j>i} |\boldsymbol{w}_i^H \boldsymbol{h}_j|^2 + 2\sigma^2 \|\boldsymbol{w}_i^H\|^2. \tag{3.13}$$

After MMSE filtering, the output can be approximated as an equivalent scalar channel

$$z_i = a_i x_i + \eta_i, \qquad a_i = \boldsymbol{w}_i^H \boldsymbol{h}_i, \qquad \eta_i \sim \mathcal{CN}(0, \sigma_{i,\text{eff}}^2). \tag{3.14}$$

For a general bit-labeled constellation alphabet $\mathcal{S}$, let $b_m(s) \in \{0, 1\}$ denote the m^{th} label bit of symbol s, and define

$$\mathcal{S}_{m,b} = \{s \in \mathcal{S} : b_m(s) = b\}, \qquad b \in \{0, 1\}. \tag{3.15}$$

In this chapter, the transmitted coded bits are assumed to be independent and equally probable. Therefore, the prior term cancels in the likelihood function. Under the scalar Gaussian approximation in (3.14), the standard likelihood-marginalization form of the soft-output demapper [9] gives the detector LLR of the m^{th} bit as

$$L_{x_i,m} = \log \frac{\sum_{s \in \mathcal{S}_{m,0}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}{\sum_{s \in \mathcal{S}_{m,1}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}, \tag{3.16}$$

For BPSK modulation, $\mathcal{S} = \{+1, -1\}$ with $0 \mapsto +1$ and $1 \mapsto -1$, and (3.16) becomes

$$L_{x_i} = \frac{4\,\text{Re}\{a_i^* z_i\}}{\sigma_{i,\text{eff}}^2}. \tag{3.17}$$

The corresponding soft symbol estimate is the posterior mean obtained from the same equally likely symbol prior [9]:

$$\hat{s}_i = \mathbb{E}[x_i | z_i] = \frac{\sum_{s \in \mathcal{S}} s \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}{\sum_{s \in \mathcal{S}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}. \tag{3.18}$$

This posterior mean reduces to

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$$\hat{s}_i = \tanh(L_{x_i}/2) \tag{3.19}$$

if BPSK modulation is used. The expression in (3.18) is therefore a detector-output soft estimate for the uncoded receiver when all transmitted bits are equally likely. The soft symbol estimate is then subtracted from the received vector:

$$\boldsymbol{y}_{(i+1)} = \boldsymbol{y}_{(i)} - \boldsymbol{h}_i \hat{s}_i. \tag{3.20}$$

The process then goes back to re-calculate the coefficient $\boldsymbol{w}_{i+1}^H$ by using (3.4) with $\boldsymbol{H}_{(i+1)}$. The LLR of x_{i+1} and its soft symbol estimate $\hat{s}_{i+1}$ can also be obtained by (3.16) and (3.18). This procedure is repeated until all users are detected.

If $\hat{s}_i$ is unreliable, the subtraction leaves a residual of cancellation in $\boldsymbol{y}_{(i+1)}$. This residual is the main source of error propagation in MMSE-SIC, and motivates the variance-aware coded receiver discussed in Chapter 4.

3.3 Uncoded Simulation Results

The main uncoded multi-resource setting uses BPSK transmission for each user, $R = 2$ resources, independent Rayleigh flat fading, and perfect CSI are assumed. The MMSE-SIC detector is compared with HDS-CDMA with ML detection [7]. We consider two large-scale fading cases: $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$. $\bar{E}_b$ in the simulations represent the average transmit energy per bit per user, normalized by the number of resources.

The results in figure. 3.1 and 3.2 show that HDS-CDMA with ML detection, which

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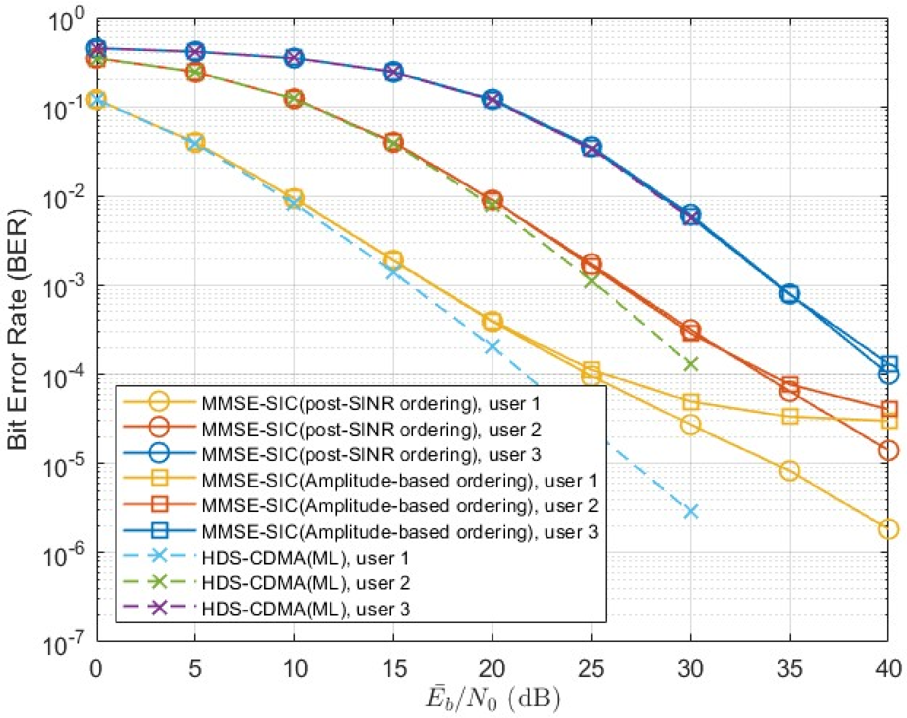

Figure 3.1: Multi-resource system. BER of MMSE-SIC and HDS-CDMA with ML detection with $R = 2, U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

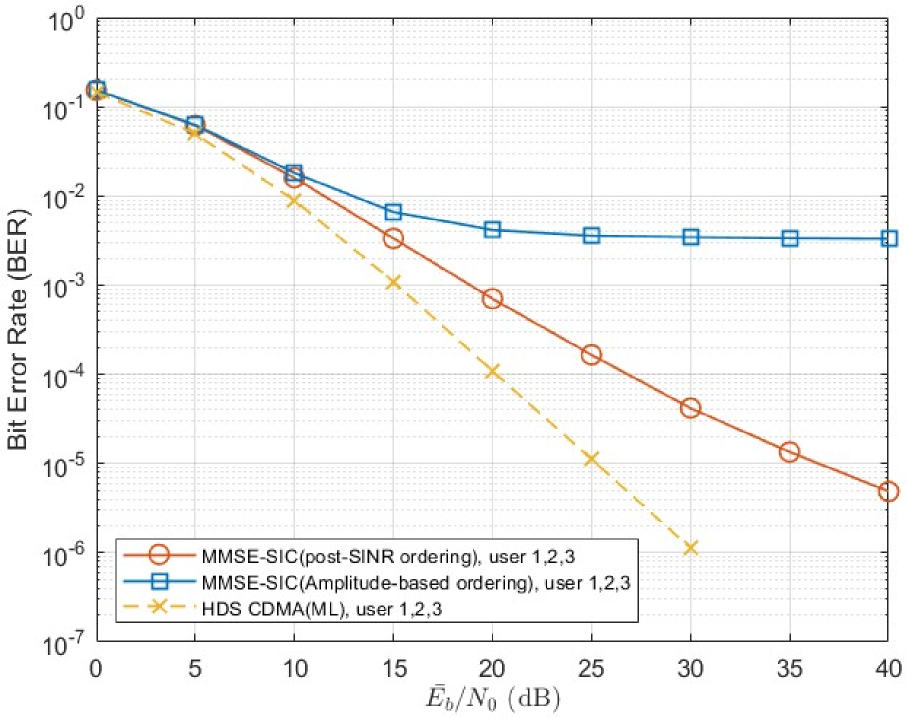

Figure 3.2: Multi-resource system. BER of MMSE-SIC and HDS-CDMA with ML detection with $R = 2, U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$.

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are indicated by the solid lines, achieves better uncoded BER than MMSE-SIC in both large-scale fading cases. For the unequal-β_k case in figure. 3.1, HDS-CDMA performs about 3 dB better than MMSE-SIC at BER $= 10^{-4}$ for user 1. For the equal-β_k case in figure. 3.2, HDS-CDMA performs about 7 dB better than MMSE-SIC at BER $= 10^{-4}$. Therefore, the performance loss of MMSE-SIC is much more significant when the users have similar average received powers.

The smaller BER gap of two detector in the unequal-β_k case is caused by the large-scale fading disparity. In particular, for the user with the smallest average received power, MMSE-SIC is close to HDS-CDMA within the simulated SNR range.

The detection ordering comparison further explains the behavior of MMSE-SIC. In the unequal-β_k case, amplitude-based ordering only causes a slight degradation around 25 dB for user 1. In the equal-β_k case, however, it degrades heavily.

These results indicate that multi-resource PD-NOMA improves user separation compared with single-resource soft SIC, but HDS-CDMA with ML detection remains the stronger uncoded benchmark. The residual performance gap thus motivates the coded system considered in the next chapter.

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Chapter 4 Coded Multi-Resource Systems with IDD

This chapter extends the multi-resource receiver design by incorporating channel coding and an iterative detection and decoding (IDD) framework. The motivation is to alleviate the error-propagation issue that arises in uncoded SIC receivers. In the coded IDD receiver, the detector exchanges soft information with the FEC decoder and refines the interference-cancellation process through repeated a priori information updates.

The chapter then investigates the block error rate (BLER) performance gap among MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD in coded multi-resource uplink NOMA systems. The same receiver concepts are further extended to multi-resource OFDM systems, where the dynamic channel conditions in uplink OFDM NOMA motivate the consideration of both synchronous and asynchronous OFDM receiver designs in the simulations.

4.1 Introduction of IDD structure

For coded systems, the detector is typically embedded in an iterative detection and decoding (IDD) receiver. In this structure, soft information is exchanged between the detector and the FEC decoder [17–19]. The detector produces bit-level LLRs for the

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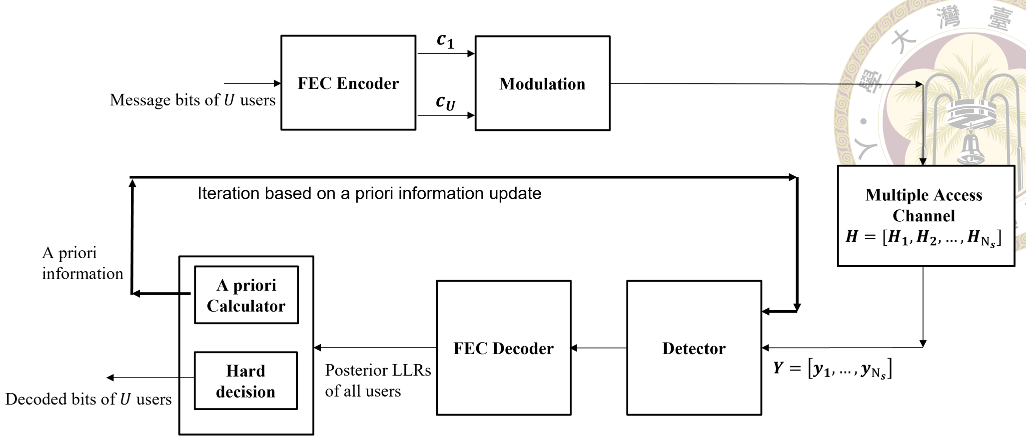

Figure 4.1: Coded NOMA system with IDD receiver

coded bits, while the decoder uses the code constraints to refine the reliability information and returns updated soft information to the detector, which is the a priori information of next detector calculation. In our simulations, the IDD receiver is implemented with a maximum number of iterations $t_{\max}$, and the final decoder output is used to make hard decisions on the transmitted bits.

4.2 Coded Multi-resource system model

In following chapters, different users are allowed to employ different modulation schemes and different channel code lengths. Let $\boldsymbol{c}_i$ denote user $i's$ transmitted codeword of length $N_{c,i}$,

$$\boldsymbol{c}_i = [c_{i,1}, c_{i,2}, \dots, c_{i,N_{c,i}}]^T, \tag{4.1}$$

After modulation, the codeword is grouped into N_s modulation symbols. At symbol time ℓ, the transmitted vector is

$$\boldsymbol{x}_\ell = [x_{1,\ell}, x_{2,\ell}, \dots, x_{U,\ell}]^T, \tag{4.2}$$

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Refer to (3.1), the received vector at symbol time ℓ is

$$\boldsymbol{y}_\ell = \boldsymbol{H}_\ell \boldsymbol{x}_\ell + \boldsymbol{n}_\ell. \tag{4.3}$$

In the block-fading simulations, $\boldsymbol{H}_1 = \boldsymbol{H}_2 = \cdots = \boldsymbol{H}_{N_s}$. For simplicity, we omit the subscript ℓ, which denotes the process is dealing with $\boldsymbol{y_\ell}$. The overall block diagram of coded NOMA system with IDD receiver is shown in figure. 4.1.

4.3 MMSE-SIC-IDD

In this section, we introduce the procedure of the MMSE-SIC-IDD receiver. The key feature of the structure is that interference cancellation in the SIC process is performed after channel decoding rather than immediately after detection.

4.3.1 Receiver Procedure

Following a similar procedure in section 3.2.1, the detection order can be obtained. For the post-SINR ordering, the additional $\bar{v}_i^{(t)}$ is considered and will be discussed in section 4.3.3. Assume the users are re-numbered from 1 to U according to this order.

For user i, the detector first computes the bit-level LLRs $\boldsymbol{L}_{\boldsymbol{c}_i}$. The decoder input LLRs are formed by adding these LLRs and the a priori information. After decoding, the decoder produces the a posteriori LLRs $\boldsymbol{L}^d_{\boldsymbol{c}_i}$, which are used to generate the decoded soft symbol estimate $\hat{\boldsymbol{s}}^d_i$. This decoded soft symbol estimate is then fed back to the MMSE-SIC detector for the next SIC stage, i.e., user $i+1$, where it is used to subtract the interference caused by user i. Intuitively, performing cancellation after decoding can improve the

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reliability of the cancelled signal and thus reduce error propagation in the SIC process.

Due to the successive decoding procedure, the extrinsic information of each user is collected asynchronously. Specifically, after decoding user i, the extrinsic LLRs are obtained as

$$\boldsymbol{L}^{e}_{\boldsymbol{c}_i} = \boldsymbol{L}^{d}_{\boldsymbol{c}_i} - \boldsymbol{L}^{in}_{\boldsymbol{c}_i}, \tag{4.4}$$

where $\boldsymbol{L}^{in}_{\boldsymbol{c}_i}$ is the input of decoder. In this way, the decoded extrinsic information of all users is provided as a priori information to the MMSE-SIC detector in the following IDD iteration.

4.3.2 Soft Information for a priori information update

Let $\mathcal{S}_i$ denote the modulation alphabet of user i, and $M_i = \log_2 |\mathcal{S}_i|$ be the number of coded bits per modulation symbol. For a symbol $s \in \mathcal{S}_i$, its binary label is written as

$$\boldsymbol{b}_i(s) = [b_{i,0}(s), b_{i,1}(s), \ldots, b_{i,M_i-1}(s)]. \tag{4.5}$$

Let $L^{a,(t)}_{c_i,\ell,m}$ denote the a priori LLR of the m^{th} coded bit in the ℓ^{th} modulation symbol of user i at a priori update iteration t, with

$$L^{a,(t)}_{c_i,\ell,m} = \log \frac{P^{a,(t)}(b_{i,\ell,m} = 0)}{P^{a,(t)}(b_{i,\ell,m} = 1)}, \qquad L^{a,(0)}_{c_i,\ell,m} = 0, \tag{4.6}$$

Where $b_{i,\ell,m}$ is the m^{th} coded bit of the ℓ^{th} modulation symbol of user i. The corresponding bit probability is

$$P^{a,(t)}(b_{i,\ell,m} = b) = \frac{1}{1 + \exp\left[-(1-2b) L^{a,(t)}_{c_i,\ell,m}\right]}, \qquad b \in \{0, 1\}. \tag{4.7}$$

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Assuming the interleaved coded bits are represented by independent bit priors at the demapper input, the a priori symbol probability is the product of its corresponding a priori bit probability [20], which is written as

$$P_{i,\ell}^{a,(t)}(s) = \prod_{m=0}^{M_i-1} P^{a,(t)}(b_{i,\ell,m} = b_{i,m}(s)). \tag{4.8}$$

The soft symbol mean used for a priori update before detection is calculated according to the a priori symbol probability [9], that is,

$$\mu_{i,\ell}^{(t)} = \mathbb{E}[x_{i,\ell} \mid \boldsymbol{L}_{c_i,\ell}^{a,(t)}] = \frac{\sum\limits_{s\in\mathcal{S}_i} sP_{i,\ell}^{a,(t)}(s)}{\sum\limits_{s\in\mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)}, \tag{4.9}$$

and the corresponding symbol variance is

$$\sigma_{i,\ell}^{2\ (t)} = \frac{\sum\limits_{s\in\mathcal{S}_i} |s|^2 P_{i,\ell}^{a,(t)}(s)}{\sum\limits_{s\in\mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)} - \left|\mu_{i,\ell}^{(t)}\right|^2. \tag{4.10}$$

For BPSK modulation with $0 \mapsto +1$ and $1 \mapsto -1$, (4.9) and (4.10) reduce to

$$\mu_{i,\ell}^{(t)} = \tanh\left(\frac{L_{c_i,\ell,0}^{a,(t)}}{2}\right), \qquad \sigma_{i,\ell}^{2\ (t)} = 1 - \left|\mu_{i,\ell}^{(t)}\right|^2. \tag{4.11}$$

For simplicity, we consider the average variance of user i over the coded block,that is,

$$\bar{v}_i^{(t)} = \frac{1}{N_s}\sum_{\ell=1}^{N_s} \sigma_{i,\ell}^{2\ (t)}, \tag{4.12}$$

The variance matrix is then defined as

$$\boldsymbol{V} = \mathrm{diag}(\bar{v}_1^{(t)}, \bar{v}_2^{(t)}, \ldots, \bar{v}_U^{(t)}), \qquad \bar{v}_i^{(t)} > 0. \tag{4.13}$$

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The soft symbol mean $\mu_{i,\ell}^{(t)}$ and variance $\bar{v}_i^{(t)}$ represent the reliability and uncertainty of the a priori information, respectively.

4.3.3 Variance-Aware MMSE Filtering

When all users' extrinsic information from the FEC decoder has been asynchronously collected, the variance-aware MMSE filter of user k [18, Eqs. (6), (7)] for next a priori based iteration is computed as

$$\tilde{\boldsymbol{w}}_k^H = \bar{v}_k^{(t)} \boldsymbol{h}_k^H \left(\boldsymbol{H} \boldsymbol{V} \boldsymbol{H}^H + 2\sigma^2 \boldsymbol{I} \right)^{-1}. \tag{4.14}$$

the corresponding post-SINR can be written as

$$\gamma_k^{\text{post}} = \frac{\left| \tilde{\boldsymbol{w}}_k^H \boldsymbol{h}_k \right|^2}{\sum_{j \neq k} \bar{v}_j^{(t)} \left| \tilde{\boldsymbol{w}}_k^H \boldsymbol{h}_j \right|^2 + 2\sigma^2 \| \tilde{\boldsymbol{w}}_k \|^2}. \tag{4.15}$$

Compared with (3.4) and (3.8), users with reliable a priori information contribute less uncertainty to the MMSE filtering process and SIC detection ordering.

4.3.4 Decoder-Aided SIC Cancellation

To simplify the expression in the following equations, we omit the subscript ℓ, which denotes that the process is dealing with $\boldsymbol{y}_\ell$. The cancellation procedure of stage 1, user 1 can benefit from the a priori information of users $2, \ldots, U$ by using

$$\boldsymbol{y}_{(1)}^{(t)} = \boldsymbol{y} - \sum_{j=2}^{U} \boldsymbol{h}_j \mu_j^{(t)}, \tag{4.16}$$

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where $\boldsymbol{y}_{(i)}^{(t)}$ is the residual vector of user i at a priori update iteration t. At SIC stage $i > 1$, the detection of user i benefits from the FEC outputs of users $1, 2, \ldots, i-1$ and from the a priori information of users $i+1, \ldots, U$. The residual vector for detecting user i is constructed as

$$\boldsymbol{y}_{(i)}^{(t)} = \boldsymbol{y} - \sum_{j=1}^{i-1} \boldsymbol{h}_j \hat{s}_j^d - \sum_{j=i+1}^{U} \boldsymbol{h}_j \mu_j^{(t)}. \tag{4.17}$$

The first summation cancels the users that have already been decoded in the current SIC pass by using the decoder-aided soft symbol estimates $\hat{s}_j^d$. This will be discussed in Section 4.3.5. The second summation cancels the users that have not yet been decoded in the current pass by using the a priori soft symbol mean $\mu_j^{(t)}$. The cancellation term in (4.17) differs from that used in the uncoded MMSE-SIC detection. In the uncoded case, the cancelled signal is directly derived from the detector output at the current SIC stage, and the corresponding residual uncertainty is not explicitly considered in MMSE filtering. In the coded IDD receiver, however, users $1, \ldots, i-1$ have already been decoded in the current SIC pass and the outputs provide more reliable soft symbol estimates. Therefore, the decoder-aided cancellation is modeled together with its residual variance $\bar{v}_c^{can}$, whose detailed calculation is given in Section 4.3.5.

Similar to the uncoded MMSE-SIC procedure in (3.11) – (3.14), the downsized channel matrix and the downsized variance matrix are used after each stage. The filter output for user i at a priori update iteration t is

$$z_i^{(t)} = \tilde{\boldsymbol{w}}_i^H \boldsymbol{y}_{(i)}^{(t)}, \tag{4.18}$$

with equivalent scalar gain

$$a_i^{(t)} = \tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_i. \tag{4.19}$$

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The effective variance of the residual interference and noise with additional residual uncertainty is

$$\sigma_{i,\text{eff}}^{2\ (t)} = \sum_{j>i} \bar{v}_j^{(t)} \left|\tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_j\right|^2 + \sum_{c<i} \bar{v}_c^{can} \left|\tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_c\right|^2 + 2\sigma^2 \left\|\tilde{\boldsymbol{w}}_i^H\right\|^2. \tag{4.20}$$

Unlike the uncoded MMSE-SIC detection, the coded receiver explicitly accounts for the residual uncertainty of decoder-aided cancellation.

4.3.5 LLR calculation and soft estimate

Define the set $\mathcal{S}_{i,m,b}$ is the subset of constellation symbols of user i whose m^{th} bit label is equal to b:

$$\mathcal{S}_{i,m,b} = \{s \in \mathcal{S}_i : b_{i,m}(s) = b\}, \qquad b \in \{0,1\}. \tag{4.21}$$

After MMSE filtering and decoder-aided cancellation, the LLR of detector output is computed from $z_i^{(t)}$, the scalar gain $a_i^{(t)}$, the effective variance $\sigma_{i,\text{eff}}^{2\ (t)}$ and the a priori LLRs in (4.6). The a priori symbol probability is given by (4.8), where the bit probabilities are obtained from $L_{c_i,\ell,m}^{a,(t)}$ by the LLR-to-probability rule in (4.7). The subscript ℓ and superscript (t) are omitted in the following scalar detection equations and LLR expressions respectively for compactness. The detector produces an extrinsic bit-level LLR for the m^{th} bit of user i, $L_{c_i,m}$ [20] by using the a priori information of the other bits in the same modulation

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symbol, but not the a priori information of the target bit itself:

$$L_{c_i,m} = \log \frac{\displaystyle\sum_{s\in\mathcal{S}_{i,m,0}} \exp\left(-\frac{\left|z_i^{(t)}-a_i^{(t)}s\right|^2}{\sigma_{i,\text{eff}}^{2\ (t)}} + \sum_{\substack{m'=0\\ m'\neq m}}^{M_i-1} \log P^{a,(t)}(b_{i,m'} = b_{i,m'}(s))\right)}{\displaystyle\sum_{s\in\mathcal{S}_{i,m,1}} \exp\left(-\frac{\left|z_i^{(t)}-a_i^{(t)}s\right|^2}{\sigma_{i,\text{eff}}^{2\ (t)}} + \sum_{\substack{m'=0\\ m'\neq m}}^{M_i-1} \log P^{a,(t)}(b_{i,m'} = b_{i,m'}(s))\right)}. \quad (4.22)$$

The input of FEC decoder for user i is the sum of the detector extrinsic bit LLR and the target-bit a priori LLR:

$$L_{c_i,m}^{\text{in}} = L_{c_i,m} + L_{c_i,m}^{a,(t)}. \quad (4.23)$$

The FEC decoder then produces posterior LLRs $\boldsymbol{L}_{\boldsymbol{c}_i}^{d}$ and will be used to form the decoder-aided soft symbol estimate for SIC cancellation. The calculation follows the same bit-probability, symbol-probability, and posterior-mean rules in (4.7), (4.8), and (4.9), with the a priori LLRs replaced by the decoder posterior LLRs. Therefore, for each symbol position ℓ, the decoded soft estimate $\hat{s}_{i,\ell}^{d}$ is obtained from $\boldsymbol{L}_{\boldsymbol{c}_i}^{d}$ in the same way that $\mu_{i,\ell}^{(t)}$ is obtained from $\boldsymbol{L}_{c_i,\ell}^{a,(t)}$.

The only additional quantity required in the decoder-aided SIC process is the cancellation residual variance $\bar{v}_i^{\text{can}}$, which is the second term of effective variance in (4.20). This quantity is computed according to $\hat{s}_{i,\ell}^{d}$ and follows (4.9)–(4.12) with the soft symbol mean replaced by the decoded soft estimate.

Summary of MMSE-SIC-IDD The overall procedure of the MMSE-SIC-IDD receiver is summarized in Algorithm 1. The number of a priori information update iterations is denoted by t_{max}, where $t_{\text{max}} = 0$ means no a priori information update is used. The block diagram of the MMSE-SIC-IDD receiver is shown in figure 4.2.

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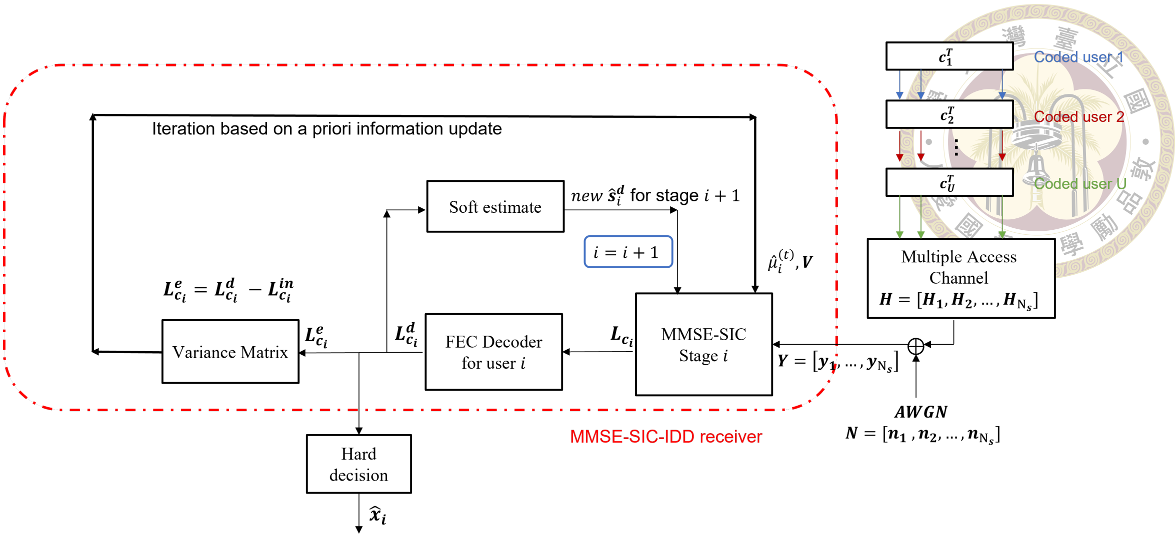

Figure 4.2: The block diagram of the MMSE-SIC-IDD receiver

Algorithm 1 MMSE-SIC-IDD receiver for one coded block

```
Initialize the a priori LLRs L^{a,(0)}_{c_i,ℓ,m} = 0 for all users, coded-symbol positions, and bit positions.
for t = 0, 1, ..., t_max do
    Compute μ^{(t)}_{i,ℓ} and σ^{2 (t)}_{i,ℓ} for all users.
    Form the variance matrix V from the block-averaged variances.
    Determine the detection order and re-number the users as 1, ..., U.
    for i = 1, 2, ..., U do
        Compute the variance-aware MMSE filter in (4.14).
        Build y^{(t)}_{(i)} using (4.16) or (4.17).
        Generate detector extrinsic LLRs using (4.22).
        Form the decoder input LLRs using (4.23).
        Decode user i and obtain ŝ^d_i and v̄^{can}_i.
    end for
    Set the next a priori LLRs using (4.4).
end for
Make hard decisions from the final decoder outputs.
```

4.4 MMSE-PIC-IDD

In this section, we introduce the MMSE-PIC-IDD receiver, which is a parallel interference cancellation (PIC) receiver that also exchanges soft information with the FEC decoder. Compared to the successive interference cancellation (SIC), PIC cancels all other users in parallel instead of successively decoding one user at a time [18]. For user k, the

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residual vector is calculated by subtracting all other users' soft symbol means:

$$\boldsymbol{y}_{k,\ell}^{(t)} = \boldsymbol{y}_\ell - \sum_{j \neq k} \boldsymbol{h}_j \mu_{j,\ell}^{(t)}. \tag{4.24}$$

The variance-aware MMSE filter follows the same form as (4.14), and the LLR calculation follows the MMSE-SIC procedure.

PIC avoids SIC ordering and reduces decoding latency because users can be processed in parallel. However, it relies directly on the a priori information of all interfering users, so the iteration of a priori information update is especially important. SIC, in contrast, performs ordered stage-by-stage cancellation and can additionally use decoded-user feedback from earlier stages in the same pass, albeit at the expense of ordering dependence and increased latency.

4.5 MAP-IDD

In this receiver, the detector follows the maximum a posteriori (MAP) rule. The detector-decoder exchange follows the same IDD structure described previously. The detector produces bit-level LLRs, the FEC decoder refines them, and the decoder extrinsic information is fed back to the detector as the a priori information for the next update.

In this section, we mainly focus on how the a priori information is operated in the MAP detector. Similar to Section 2.3, the MAP detector considers all possible superimposed signals. For simplicity, the subscript ℓ is omitted in this section, which means that the receiver is processing the ℓth transmission symbol. The possible superimposed signal set is written as

$$\mathcal{S} = \left\{ \boldsymbol{s}_0, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_{N_L - 1} \right\}, \tag{4.25}$$

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where

$$N_L = \prod_{k=1}^{U} |\mathcal{S}_k| = 2^{\sum_{k=1}^{U} M_k}. \tag{4.26}$$

The l^{th} possible superimposed signal is generated by one combination of all users' modulation symbols. Let this combination be

$$\boldsymbol{x}_l = [x_{1,l}, x_{2,l}, \ldots, x_{U,l}]^T, \qquad x_{k,l} \in \mathcal{S}_k. \tag{4.27}$$

Then the corresponding superimposed signal is

$$\boldsymbol{s}_l = \boldsymbol{H}\boldsymbol{x}_l. \tag{4.28}$$

For example, consider three users employing BPSK, QPSK, and 16QAM, respectively. In this case,

$$|\mathcal{S}_1| = 2, \qquad |\mathcal{S}_2| = 4, \qquad |\mathcal{S}_3| = 16,$$

and the MAP detector needs to consider

$$N_L = 2 \times 4 \times 16 = 128$$

possible superimposed signals for each received symbol. One possible combination is, for example:

$$\boldsymbol{x}_l = [x_{1,l}, x_{2,l}, x_{3,l}]^T = \left[+1, \frac{1+j}{\sqrt{2}}, \frac{1+3j}{\sqrt{10}}\right]^T, \tag{4.29}$$

where the first entry is selected from the BPSK constellation, the second entry is selected from the QPSK constellation, and the third entry is selected from the 16QAM constella-

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tion. This symbol combination produces one possible superimposed signal

$$\boldsymbol{s}_l = \boldsymbol{H}\left[+1, \frac{1+j}{\sqrt{2}}, \frac{1+3j}{\sqrt{10}}\right]^T. \tag{4.30}$$

The MAP detector repeats this construction for all possible symbol combinations.

Without a priori information, all possible superimposed signals are assumed to be equally likely. In MAP-IDD, however, the detector uses the a priori LLRs from the previous IDD update to weight each possible superimposed signal. From (4.7), the a priori probability of the l^{th} possible superimposed signal is calculated as the product of the a priori probabilities of all its corresponding coded bits:

$$P_l^{a,(t)} = \prod_{k=1}^{U} \prod_{m=0}^{M_k-1} P^{a,(t)}\left(b_{k,m} = b_{k,m}(x_{k,l})\right). \tag{4.31}$$

Therefore, the APP of the l^{th} possible superimposed signal is proportional to

$$P^{(t)}(\boldsymbol{s}_l|\boldsymbol{y}) \propto \exp\left(-\frac{\|\boldsymbol{y}-\boldsymbol{s}_l\|^2}{2\sigma^2}\right) P_l^{a,(t)}. \tag{4.32}$$

(4.32) shows that the a priori information changes the probability of each possible superimposed signal according to its bit labels.

For the m^{th} bit of user k, the detector output LLR is obtained by summing the APPs of all possible superimposed signals whose corresponding bit is equal to 0 or 1:

$$L_{c_k,m}^{\text{MAP}} = \log \frac{\sum\limits_{l:b_{k,m}(x_{k,l})=0} P^{(t)}(\boldsymbol{s}_l|\boldsymbol{y})}{\sum\limits_{l:b_{k,m}(x_{k,l})=1} P^{(t)}(\boldsymbol{s}_l|\boldsymbol{y})}. \tag{4.33}$$

After the detector LLRs are derived synchronously for all users, the subsequent IDD procedure follows the same steps described previously.

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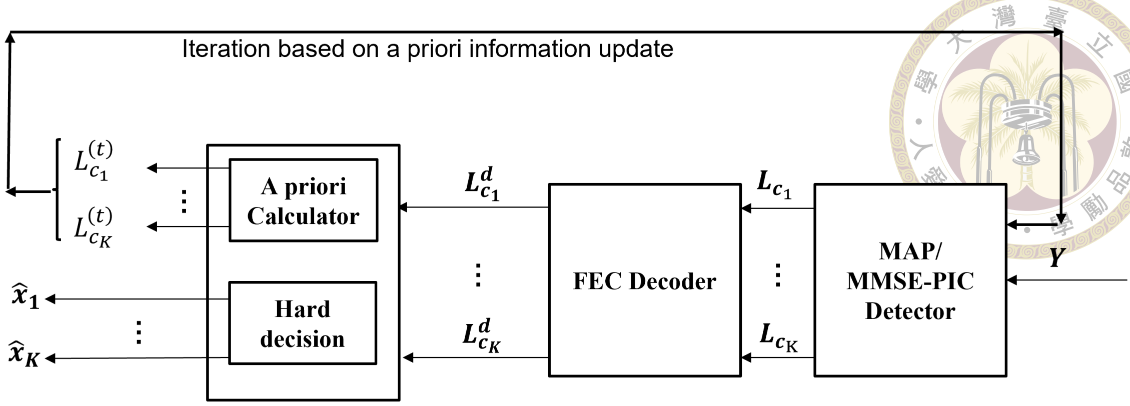

Figure 4.3: The block diagram of the MAP-IDD/MMSE-PIC-IDD receiver

MAP-IDD provides a strong coded joint-detection ability because it evaluates the APPs over all possible superimposed signals. However, its complexity increases with $N_L = \prod_{k=1}^{U} |\mathcal{S}_k|$, and therefore grows rapidly as the number of users or the number of coded bits per modulation symbol increases.

Summary of MAP-IDD/MMSE-PIC-IDD The block diagram of the MAP-IDD/MMSE-PIC-IDD receiver is shown in figure 4.3.

4.6 Coded Simulation Results

For the MMSE-SIC-IDD receiver, post-SINR ordering is used. In the four-user case and the scenarios with more than four users, dynamic post-SINR ordering is considered. In this ordering strategy, the detection order is updated after each SIC stage. Accordingly, the post-detection SINR in (4.15) is recalculated at every SIC stage based on the remaining undecoded users.

We compare the block error rate (BLER) of the three receiver structures mentioned above. The simulations employ a $(1008, 504)$ LDPC code over an independent block-fading channel for every user on each resource. Two resources are used in our system and

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BPSK modulation is used for each user unless otherwise specified.

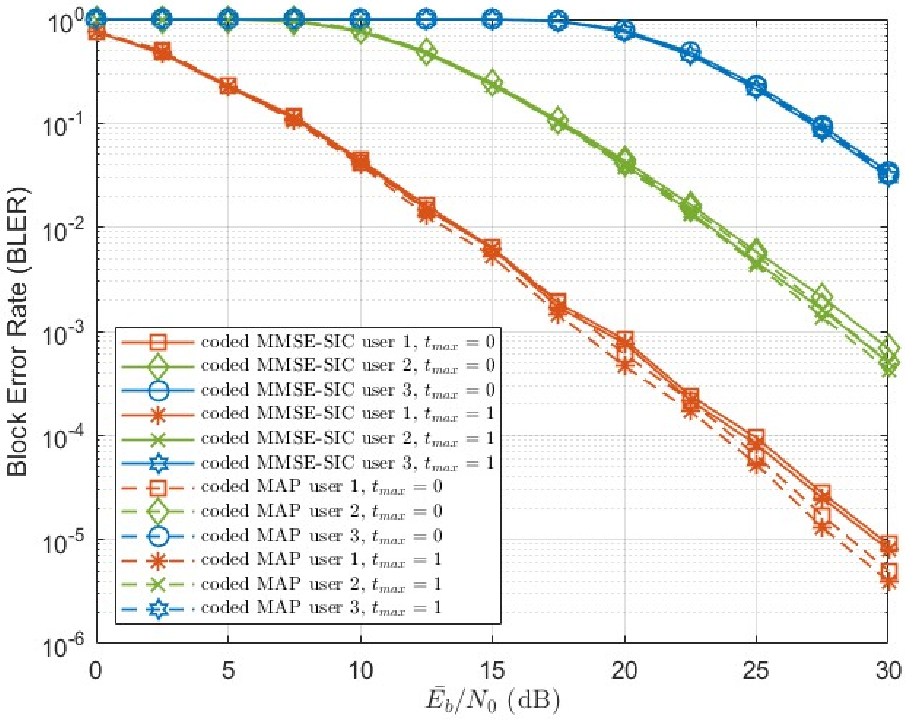

Figure 4.4: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$, $t_{\text{max}} = 0$ and $t_{\text{max}} = 1$.

Figure. 4.4 and 4.5 show the BLER performance of MMSE-SIC-IDD, MAP-IDD and MMSE-PIC-IDD with unequal-β_k case, $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$. Figure. 4.4 compares MMSE-SIC-IDD and MAP-IDD with $t_{\text{max}} = 0$ and $t_{\text{max}} = 1$. Figure. 4.5 shows the MMSE-PIC-IDD with different values of t_{max}. Compared with MMSE-SIC-IDD and MAP-IDD, the performance is close for all users, indicating that decoder-aided soft cancellation effectively reduces residual interference. In figure. 4.5, it's observed that MMSE-PIC-IDD strongly relies on reliable a priori information update because parallel cancellation relies directly on the a priori-based soft means of all interfering users.

Figure. 4.6 considers the equal-β_k case of three receivers. For the MMSE-PIC-IDD scheme, convergence of the BLER is nearly achieved by $t_{\text{max}} = 3$. Consequently, the remaining simulations in this chapter are conducted solely using $t_{\text{max}} = 3$. Compared

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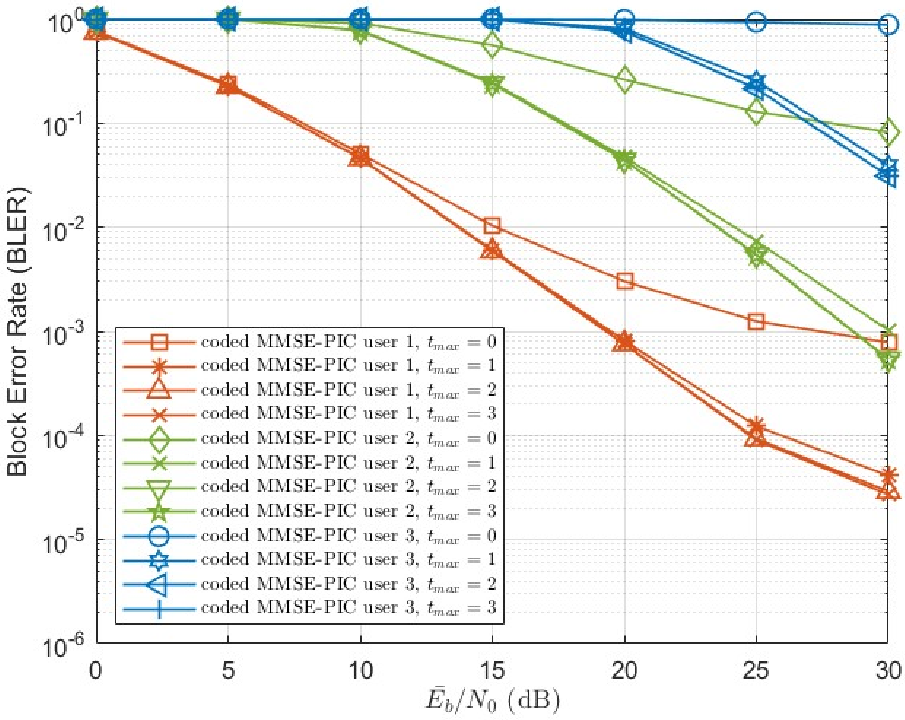

Figure 4.5: BLER of MMSE-PIC-IDD with $R = 2$, $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$, $t_{\max} = 0, 1, 2, 3$.

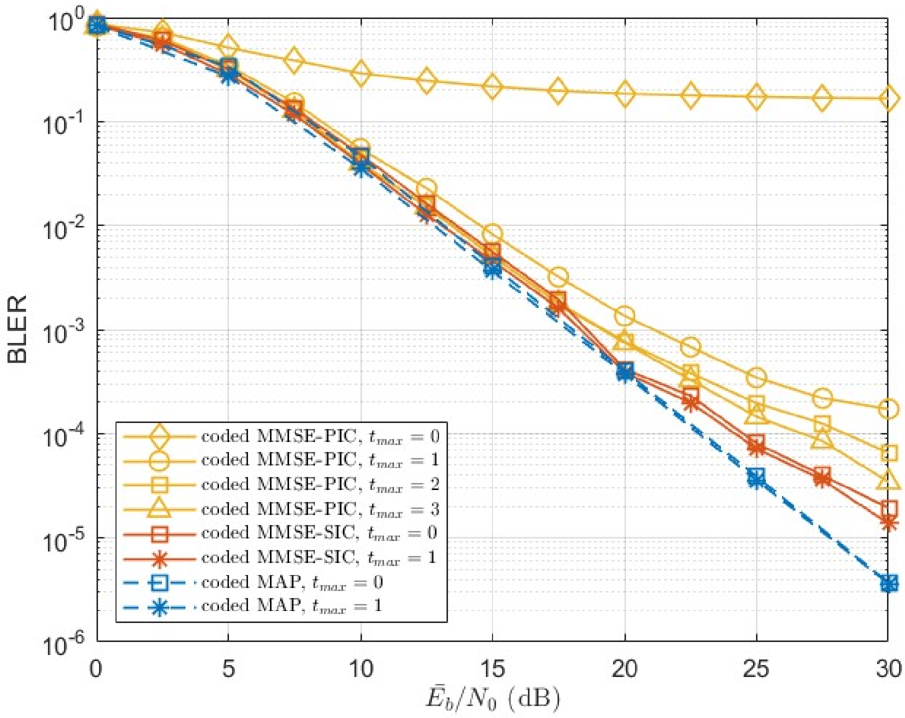

Figure 4.6: BLER of different receiver structures with $R = 2$, $U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$.

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with MMSE-SIC-IDD and MAP-IDD, the performance is also close, while MAP-IDD is slightly better in the high-SNR region. It can also be observed that for both unequal-β_k case and equal-β_k case, the gain from increasing the value of t is limited. This is due to the assumption of a block fading channel and the relatively moderate system loading.

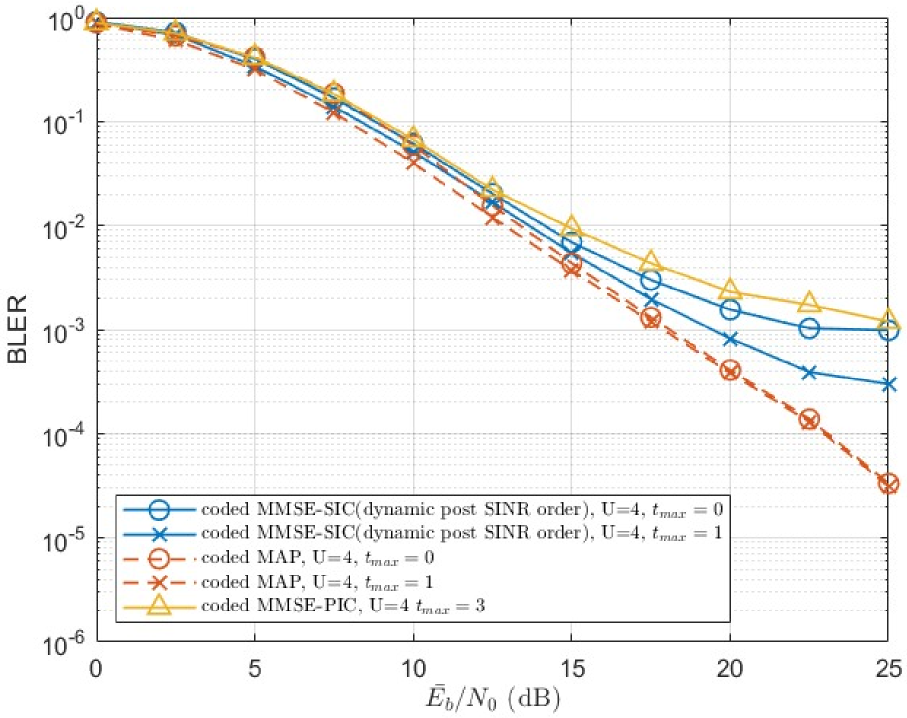

Figure 4.7: BLER of different receiver structures with $R=2$, $U=4$ and $\beta_k=1$.

Figure. 4.7 and 4.8 are the cases of four-user and six-user systems with equal-β_k values, respectively. It can be seen that MAP-IDD maintains superior BLER performance owing to its near-optimal detection capability, whereas MMSE-based receivers become more sensitive to the increase in the number of users.

Figure. 4.9 and 4.10 consider three users using modulation (BPSK, QPSK, 16QAM). Post-SINR ordering is used for MMSE-SIC-IDD. These two figures represent $t_{\max}=0$ and $t_{\max}=1$, respectively. Compared with BPSK modulation for every user in figure. 4.6, MAP-IDD can benefit from a priori information update even under block fading channel, since decoder feedback helps refine the APP weights of possible superimposed signals.

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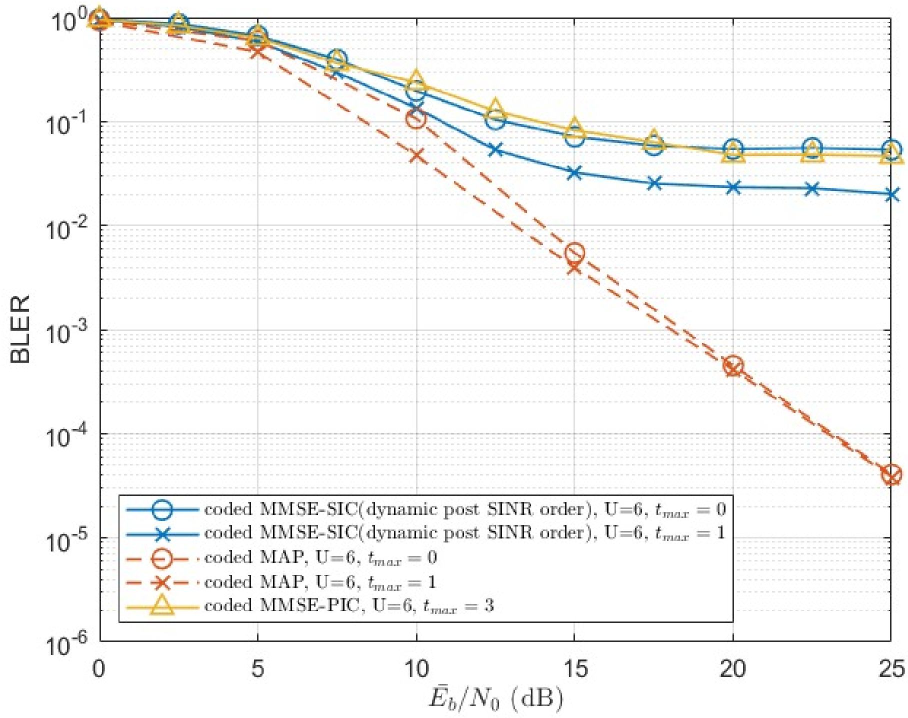

Figure 4.8: BLER of different receiver structures with $R = 2$, $U = 6$ and $\beta_k = 1$.

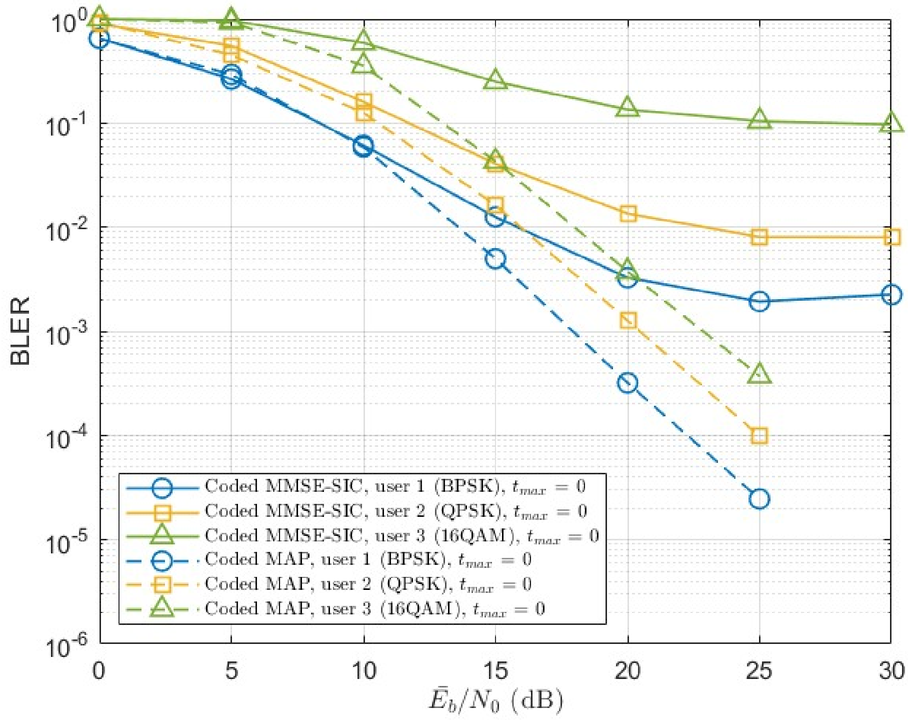

Figure 4.9: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $U = 3$ and $\beta_k = 1$. (BPSK, QPSK, 16QAM) is used. $t_{\text{max}} = 0$

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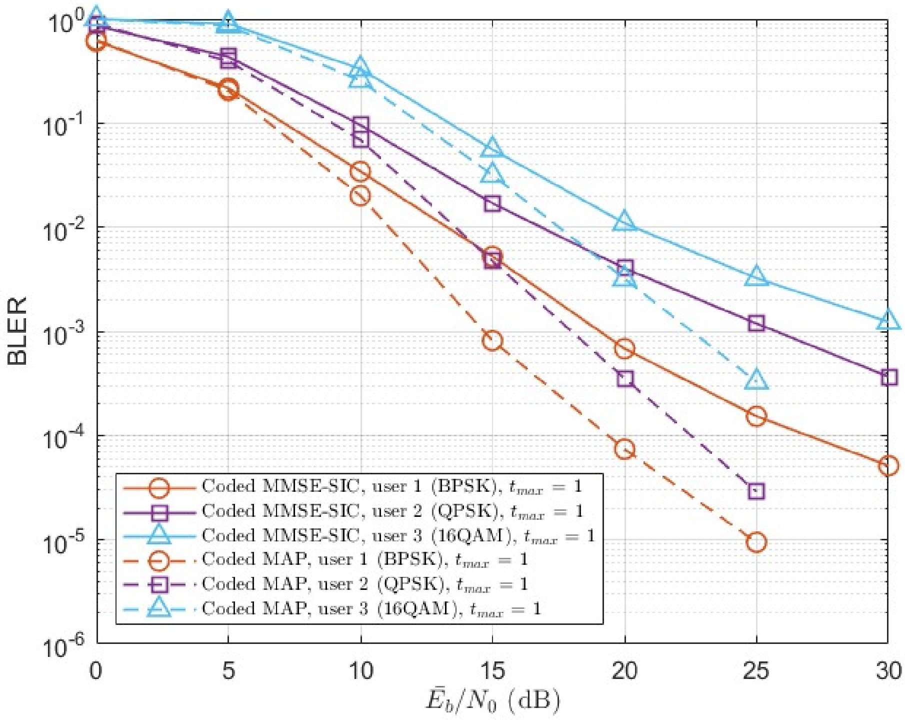

Figure 4.10: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $U = 3$ and $\beta_k = 1$. (BPSK, QPSK, 16QAM) is used. $t_{\max} = 1$

4.7 Coded multi-resource NOMA for OFDM system

This section extends the coded multi-resource receiver to an Orthogonal Frequency-Division Multiplexing (OFDM) system. Since practical broadband channels are frequency selective, OFDM is introduced to convert the multi-path channel into parallel frequency-domain observations. Furthermore, the high spectral efficiency of OFDM, together with its resilience to channel distortions, has led to its widespread adoption in practical systems such as LTE, Wi-Fi, and digital broadcasting standards [21].

We first evaluate the BLER performance of the synchronous OFDM system with different receivers discussed in above. However, perfect synchronous transmission is impractical especially for uplink transmissions, where users may arrive at the base station with different timing offsets [22]. Therefore, this section further extends the study to asynchronous uplink transmission and examines the robustness of these receivers under user-dependent timing offsets.

4.7.1 Coded Multi-Resource OFDM Signal Model

Related coded NOMA OFDM systems have been studied for robust multiuser transmission to enhance data rate [14, 23, 24]. Practical uplink systems often employ OFDM to handle frequency-selective fading and multipath interference. Consider a coded OFDM system with N_{FFT} subcarriers and cyclic prefix length N_{CP}. Let N_c denote the coded block length and M denote the number of coded bits per modulation symbol. Each coded block occupies

$$N_{\text{sym}} = \frac{N_c}{M N_{\text{FFT}}} \tag{4.34}$$

OFDM symbols.

After CP removal and FFT, the received signal on subcarrier q of OFDM symbol j is

$$\boldsymbol{y}_{q,j} = \boldsymbol{H}_q \boldsymbol{x}_{q,j} + \boldsymbol{n}_{q,j}. \tag{4.35}$$

where $q = 0, 1, \ldots, N_{\text{FFT}} - 1$ and $j = 0, 1, \ldots, N_{\text{sym}} - 1$. Here, $\boldsymbol{y}_{q,j} \in \mathbb{C}^{R\times 1}$ is the received vector over R resources, $\boldsymbol{H}_q \in \mathbb{C}^{R\times U}$ represents the channel vector on subcarrier q, $\boldsymbol{x}_{q,j} \in \mathbb{C}^{U\times 1}$ is the transmitted symbol vector of U users, and $\boldsymbol{n}_{q,j} \in \mathbb{C}^{R\times 1}$ denotes complex AWGN. For each subcarrier and OFDM symbol, the system follows the same multi-resource vector model as (3.1).

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4.7.2 MMSE-SIC-IDD for OFDM system

In practical wireless systems, the coded-OFDM is commonly implemented where the LLRs on subcarriers are calculated and then sent to the decoder in a block manner [25]. Therefore, we calculate the post-SINR(or the received-power metric) of each subcarrier q of user k and decide a detection order of whole coded block. The detection can be determined from an average reliability metric:

$$\bar{\Gamma}_k = \frac{1}{N_{\text{FFT}}} \sum_q \Gamma_{k,q}, \tag{4.36}$$

where $\Gamma_{k,q}$ follows the calculation in section 3.2.1.

4.7.3 Synchronous OFDM Simulation Results

We first compare MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD under the synchronous OFDM system. Unless otherwise specified, the simulations consider a two-resource system with BPSK modulation for each user with $\beta_k = 1$. $(1008, 504)$ LDPC code is used. The OFDM parameters are $N_{\text{FFT}} = 16$ and $N_{\text{CP}} = 4$. The channel model involves a five-tap quasi-Rayleigh fading channel with an exponential power delay profile in the taps for every user and every resource.

Figure. 4.11 shows the impact of the a priori information update on MMSE-PIC-IDD for the three-user case. As t_{max} increases, the decoder feedback provides more reliable soft information, and the BLER improves significantly. The result shows that MMSE-PIC-IDD nearly converges after $t_{\text{max}} = 2$.

Figure. 4.12 compares the two detection-order methods for MMSE-SIC-IDD. Both

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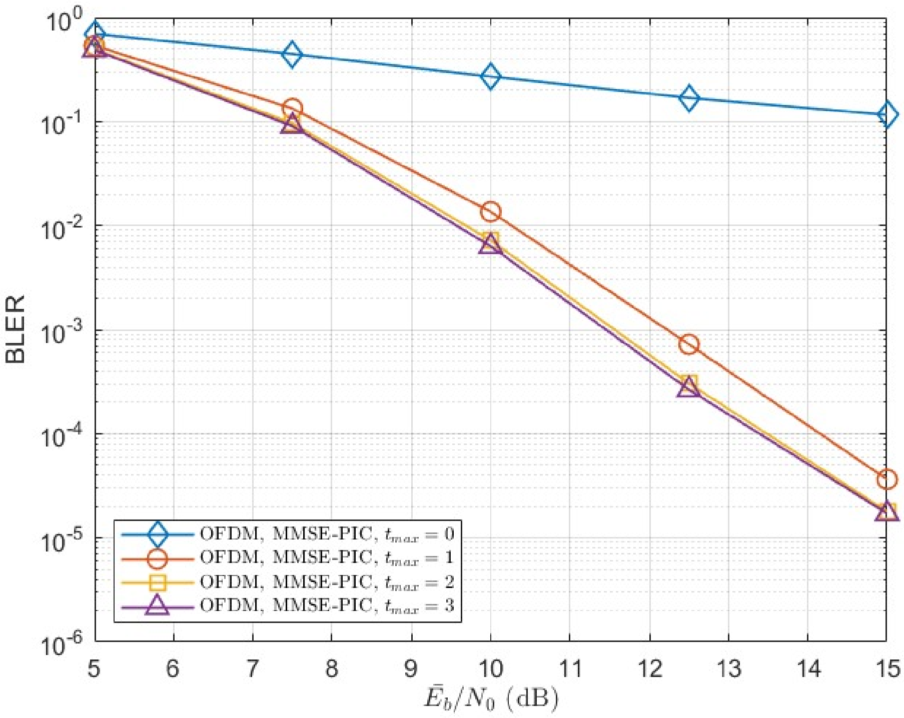

Figure 4.11: BLER of OFDM system with MMSE-PIC-IDD receiver. $R = 2$, $U = 3$, $t_{\text{max}} = 0, 1, 2, 3$.

of them benefit from the a priori information update, especially in the high-SNR region.

Figure. 4.13 compares the best-performing receiver settings for the three-user OFDM system. MAP-IDD adopts the optimal MAP detection. However, this does not necessarily mean that the overall receiver achieves the best BLER performance in coded OFDM setting, since the final performance also depends on how decoder feedback, soft information update, and residual inter-user interference (IUI) are handled across the IDD process. In this moderate-loading case, MMSE-SIC-IDD with post-SINR ordering and $t_{\text{max}} = 1$ slightly outperforms MAP-IDD. This indicates that decoder-aided cancellation, soft information update, and subcarrier channel diversity can improve the reliability of IUI suppression. On the other hand, MMSE-SIC-IDD with amplitude-based ordering is slightly worse because the ordering metric does not directly reflect the post-filtering reliability.

Figures 4.14 and 4.15 show the BLER performance with different numbers of users

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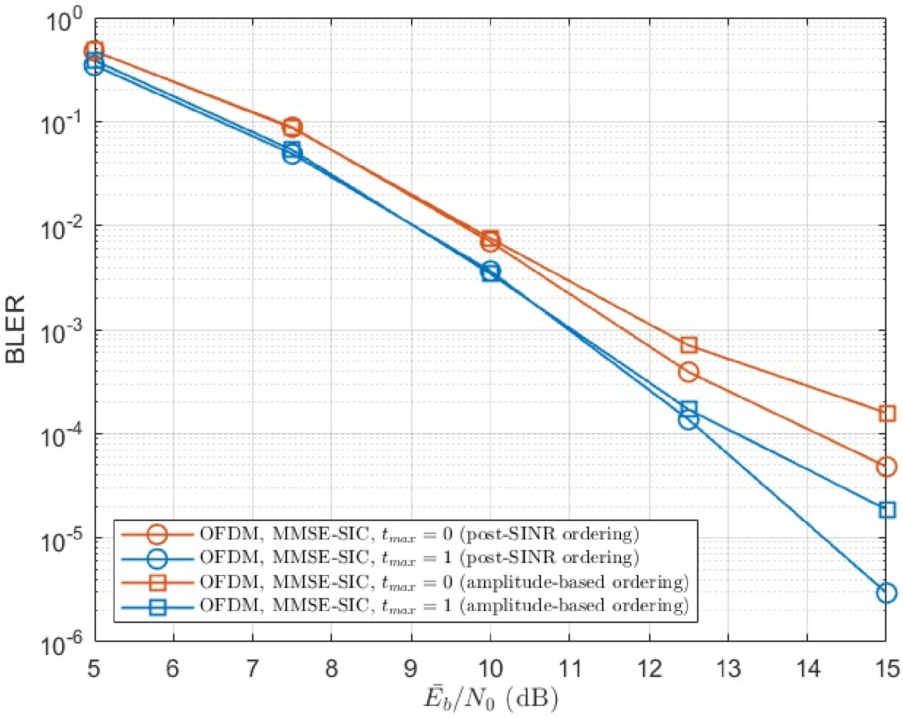

Figure 4.12: BLER of OFDM system with MMSE-SIC-IDD receiver using two detection-order methods. $R = 2, U = 3, t_{\text{max}} = 0, 1$.

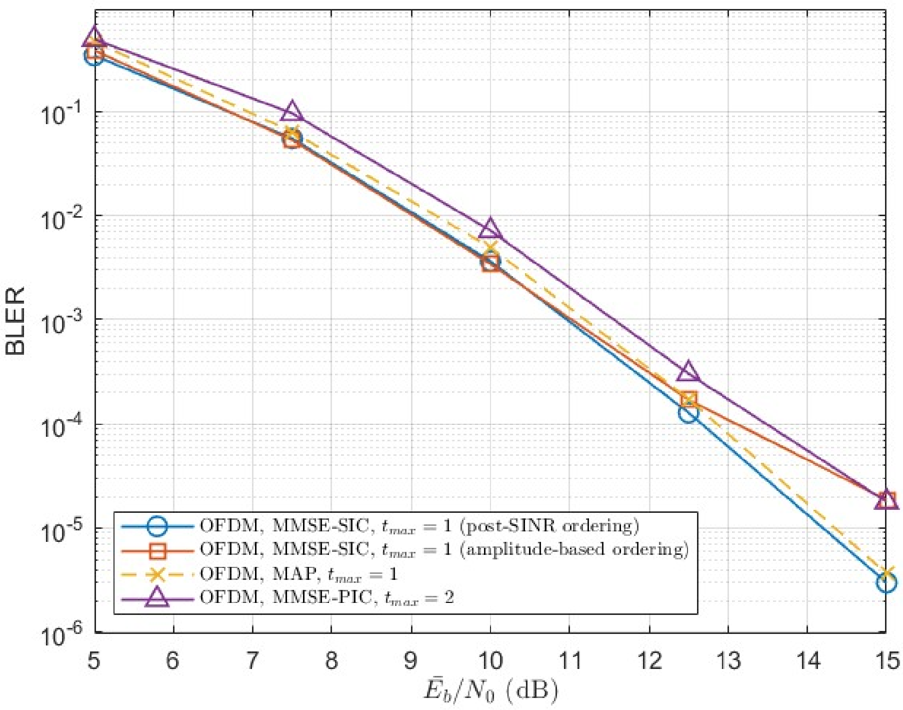

Figure 4.13: BLER comparison of MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD for OFDM system. $R = 2, U = 3$.

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for $t_{\max} = 0$ and $t_{\max} = 1$, respectively. When $t_{\max} = 0$, MMSE-SIC-IDD becomes increasingly sensitive to error propagation as the number of users increases. When $t_{\max} = 1$, decoder feedback provides more reliable soft information, and MMSE-SIC-IDD is significantly improved. In the four-user case, MMSE-SIC-IDD can approach MAP-IDD. However, the BLER degradation of MMSE-SIC-IDD becomes obvious and the error floor remains as the number of users further increases. In contrast, MAP-IDD maintains favorable BLER performance even for the six-user case. Moreover, the gain from the a priori information update becomes more pronounced as the number of users increases, which indicates that MAP-IDD can effectively exploit a priori information update when the number of possible superimposed signals becomes larger.

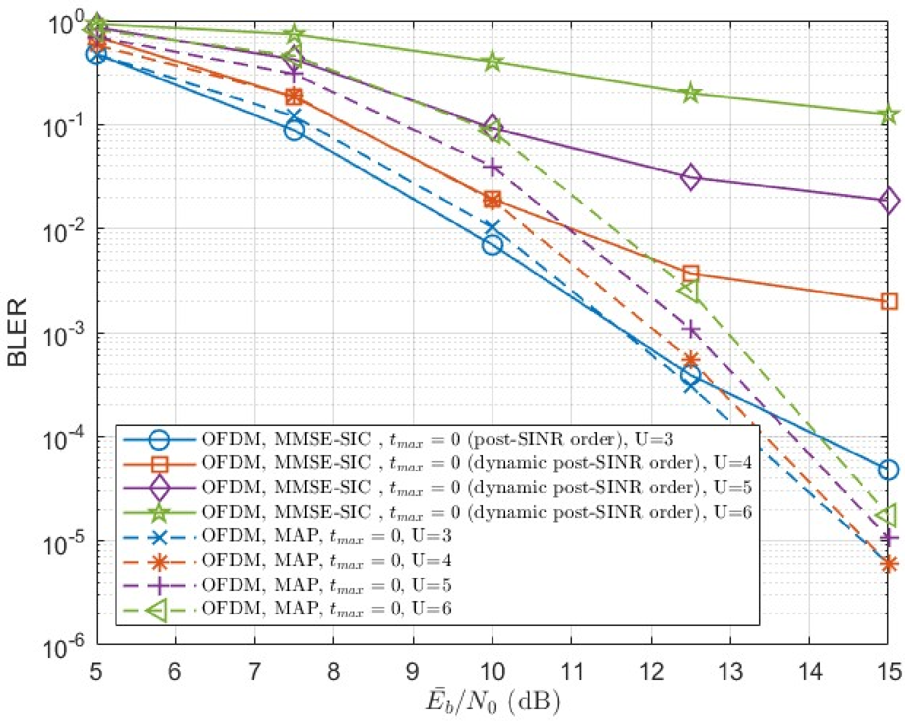

Figure 4.14: BLER comparison of MMSE-SIC-IDD and MAP-IDD with different numbers of users. $R = 2$, $U = 3, 4, 5, 6$, $t_{\max} = 0$.

Figure. 4.16 and 4.17 consider three users using modulation (BPSK, QPSK, 16QAM). Post-SINR ordering is adopted for MMSE-SIC-IDD. The two figures correspond to $t_{\max} =$

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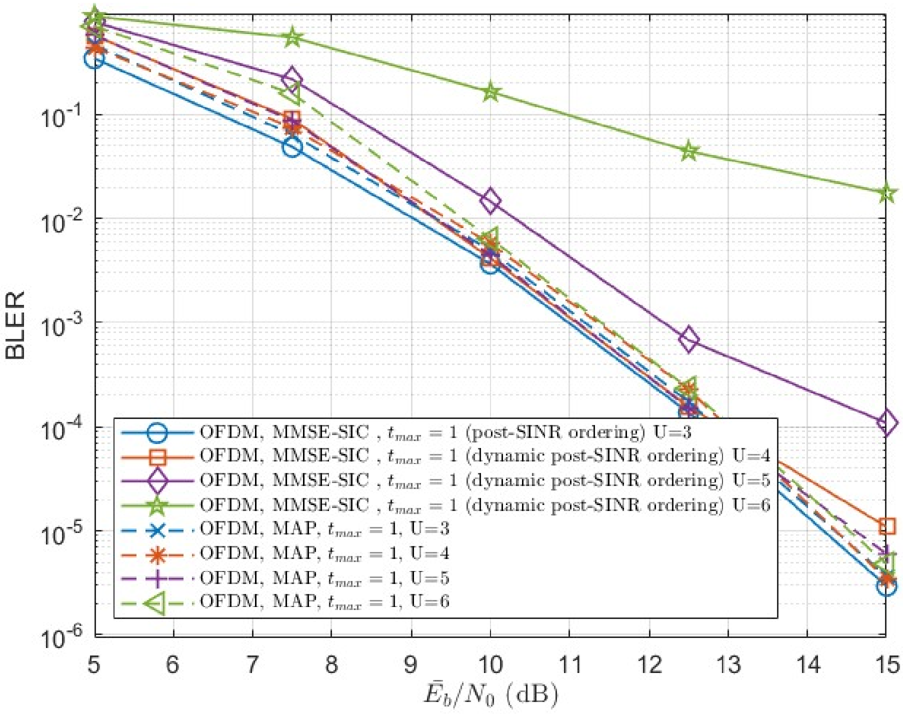

Figure 4.15: BLER comparison of MMSE-SIC-IDD and MAP-IDD with different numbers of users. $R = 2$, $U = 3, 4, 5, 6$, $t_{max} = 1$.

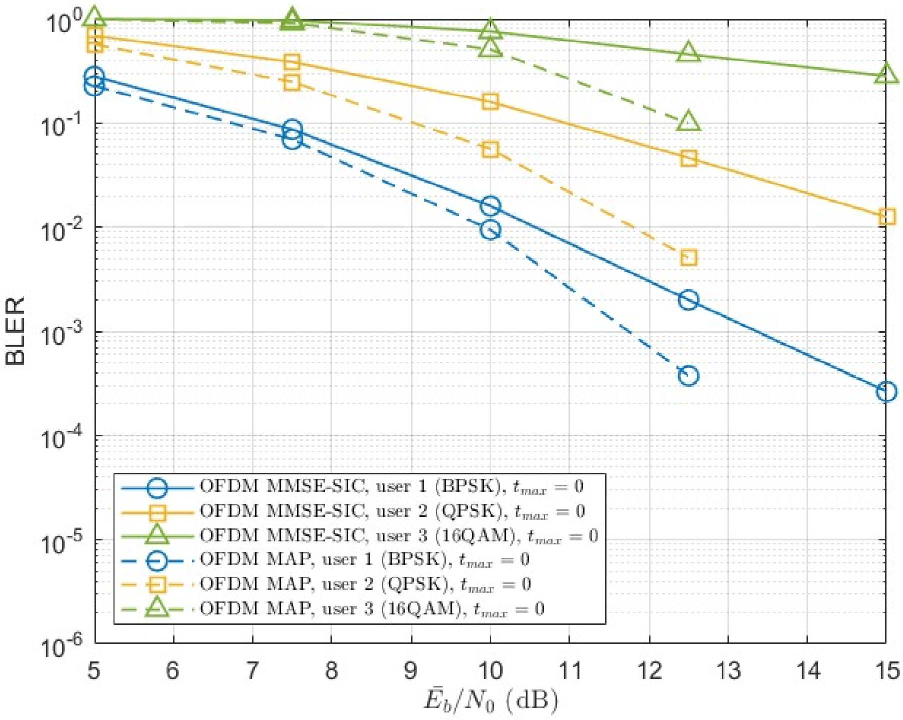

Figure 4.16: BLER of OFDM system with MMSE-SIC-IDD and MAP-IDD. $R = 2$, $U = 3$. (BPSK, QPSK, 16QAM) is used. $t_{\max} = 0$

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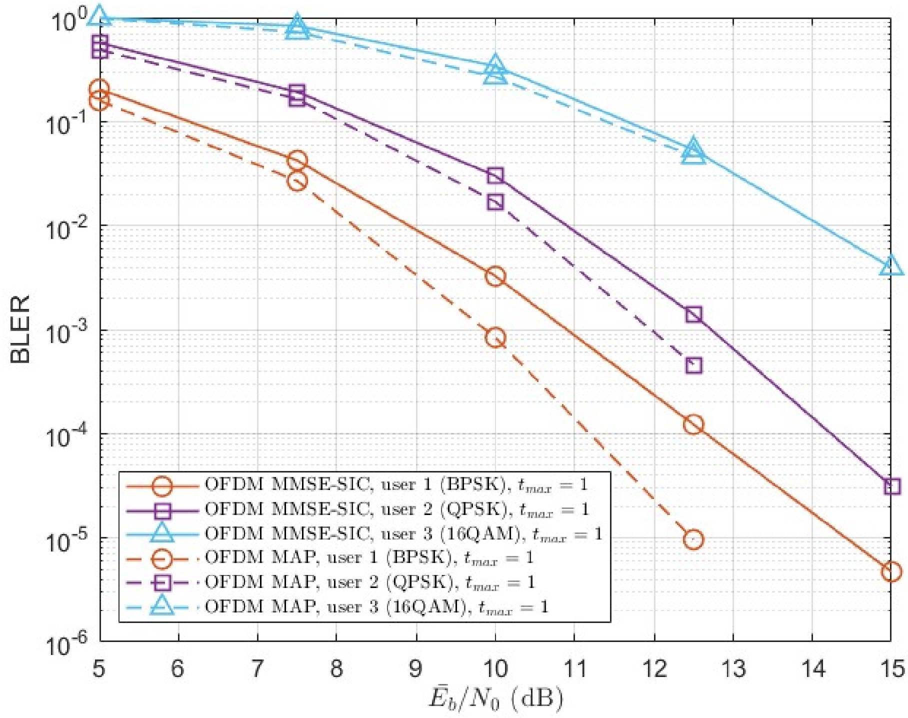

Figure 4.17: BLER of OFDM system with MMSE-SIC-IDD and MAP-IDD. $R = 2$, $U = 3$. (BPSK, QPSK, 16QAM) is used. $t_{\max} = 1$

0 and $t_{\max} = 1$, respectively. Without a priori update, MAP-IDD outperforms MMSE-SIC-IDD for all three users. When $t_{\max} = 1$, the BLER performance of both receivers is significantly improved, while MAP-IDD still performs better. The performance gap is relatively small for the 16QAM user, although this user exhibits the highest BLER because of its larger number of coded bits per modulation symbol.

4.7.4 Asynchronous OFDM NOMA system

This section extends the coded OFDM NOMA system to asynchronous uplink transmission. In practical uplink transmissions, users may arrive at the BS with different timing offsets because of propagation delay, dynamic mobile environments, or limited timing-advance precision [11, 22, 26]. If the sum of the timing offset and the maximum channel delay spread remains within the cyclic prefix (CP), the OFDM subcarriers remain orthogo-

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nal and the timing offset only introduces a phase rotation in each subcarrier symbol [27]. Therefore, a timing offset within the CP can be represented as a subcarrier-dependent phase rotation, but the multiuser detector observes different effective channel responses for different users.

In most related studies, timing offset is treated as an impairment and the receiver attempts to estimate, compensate, or cancel it [11, 22, 27]. In contrast, timing offset can also be intentionally introduced into NOMA to decrease IUI and enhance performance [28]. In this section, we evaluate the tolerance of the considered receivers when timing offsets exist in the received signals.

4.7.4.1 Asynchronous Signal Model

For a NOMA system, different users may arrive at the BS with different timing offsets. Therefore, the effective channel observed by the multiuser detector is changed, and additional multiple-access interference (MAI) or inter-symbol interference (ISI) may appear depending on the receiver alignment and the CP margin. [11]. Let τ_k denote the timing offset of user k and set user 1 as the receiver timing reference, i.e., $\tau_1 = 0$. The relative timing offset of user k is defined as

$$\Delta_k = \tau_k - \tau_1. \tag{4.37}$$

For a multi-path OFDM channel, the timing offset can be represented only as a subcarrier-dependent phase rotation when the delayed channel impulse response is still fully contained within the CP [27]. Let τ_{max} denote the maximum channel delay spread. In the five-tap channel model considered in this chapter, $\tau_{\text{max}} = 4$. Therefore, the suffi-

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cient condition for preserving the circular convolution property of OFDM is

$$\Delta_k + \tau_{\text{max}} \leq N_{\text{CP}}. \tag{4.38}$$

For the multi-resource vector model, the asynchronous effective channel matrix on subcarrier q can be written as

$$\tilde{\boldsymbol{H}}_q = \boldsymbol{H}_q \boldsymbol{\Phi}_q, \qquad \tilde{\boldsymbol{H}}_q \in \mathbb{C}^{R \times U} \tag{4.39}$$

where

$$\boldsymbol{\Phi}_q = \text{diag}\left(e^{-j2\pi q \Delta_1 / N_{\text{FFT}}}, e^{-j2\pi q \Delta_2 / N_{\text{FFT}}}, \ldots, e^{-j2\pi q \Delta_U / N_{\text{FFT}}}\right). \tag{4.40}$$

The received signal on subcarrier q of OFDM symbol j is then modeled as

$$\boldsymbol{y}_{q,j} = \tilde{\boldsymbol{H}}_q \boldsymbol{x}_{q,j} + \boldsymbol{n}_{q,j}. \tag{4.41}$$

The above model treats asynchronous transmission as a user-dependent effective channel variation. This model is exact when (4.38) holds. However, when the timing offset and the multipath delay spread exceed the CP margin, the received OFDM symbol can no longer be described only by the phase-rotated channel matrix in (4.39).

In the considered asynchronous simulations, the detectors still process the received signal through the effective phase-rotated channel model in (4.39). Therefore, when (4.38) is violated, ISI is occurred and this model should be interpreted as a receiver-side approximation. The purpose of the section is to evaluate the robustness of MAP-IDD, MMSE-SIC-IDD, and MMSE-PIC-IDD under user-dependent timing offsets.

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4.7.4.2 Asynchronous Simulation Results

The asynchronous simulations consider the two-resource systems with three-user and four-user cases. The receiver is aligned to user 1, and the relative timing offset of the other users is randomly selected from $[0, \Delta_{\max}]$. Therefore, $\Delta_{\max} = 4$ indicates that the timing offset itself is bounded by the CP length, while $\Delta_{\max} = 5$ represents a more severe asynchronous case. The OFDM parameters and the channel follow the same settings as the previous synchronous OFDM simulations. In our simulations, we consider $t_{\max} = 0$ and $t_{\max} = 1$ for MMSE-SIC-IDD and MAP-IDD, $t_{\max} = 2$ for MMSE-PIC-IDD.

Figures 4.18 and 4.19 show the BLER performance of the three-user under asynchronous OFDM system with $\Delta_{\max} = 4$. It can be seen that MMSE-SIC-IDD and MMSE-PIC-IDD remain competitive under asynchronous transmission. When $t_{\max} = 0$, MAP-IDD is slightly better in the high-SNR region. After one a priori information update shown in Figure 4.19, MMSE-SIC-IDD with two detection order strategies both improve and become slightly better than MAP-IDD over the simulated SNR range. This indicates that MMSE-based detection with a priori information update is relatively robust to asynchronous effects because they treat the undecoded users as interference-plus-noise. Therefore, part of the asynchronous mismatch can be absorbed into the effective interference-plus-noise term.

Figures 4.20 and 4.21 show the corresponding results when $\Delta_{\max} = 5$. Compared with $\Delta_{\max} = 4$, the overall BLER becomes worse because the asynchronous effect is stronger. A similar trend can be observed compared with $\Delta_{\max} = 4$. In particular, Figure 4.20 shows that even without a priori information update, the two MMSE-SIC-IDD ordering methods can slightly outperform MAP-IDD, which suggests that MAP-IDD is

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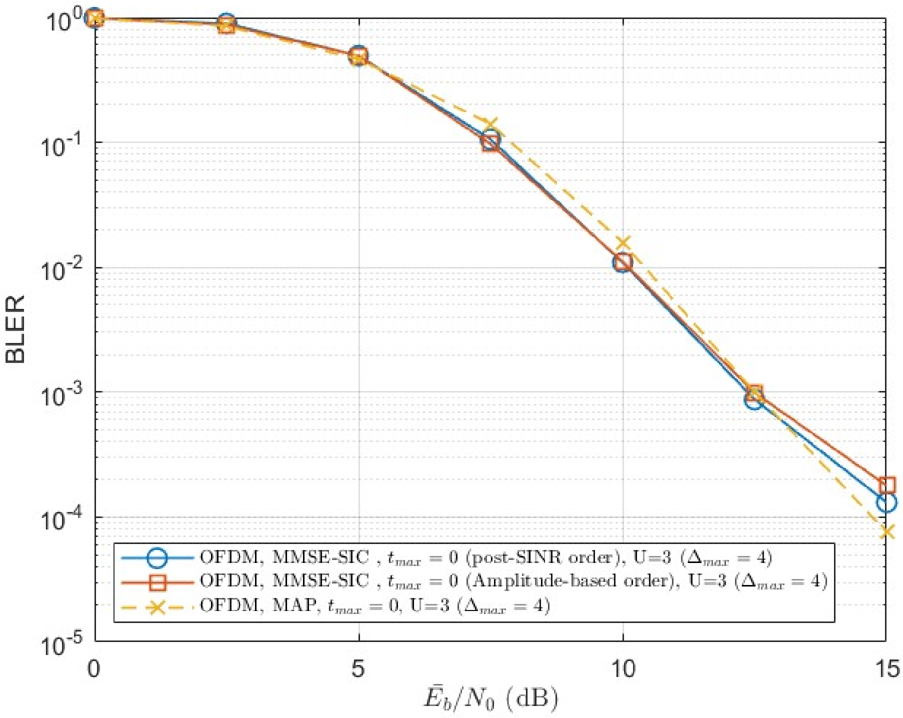

Figure 4.18: BLER of asynchronous OFDM system with MMSE-SIC-IDD and MAP-IDD. $R = 2, U = 3, \Delta_{\max} = 4, t_{\max} = 0$.

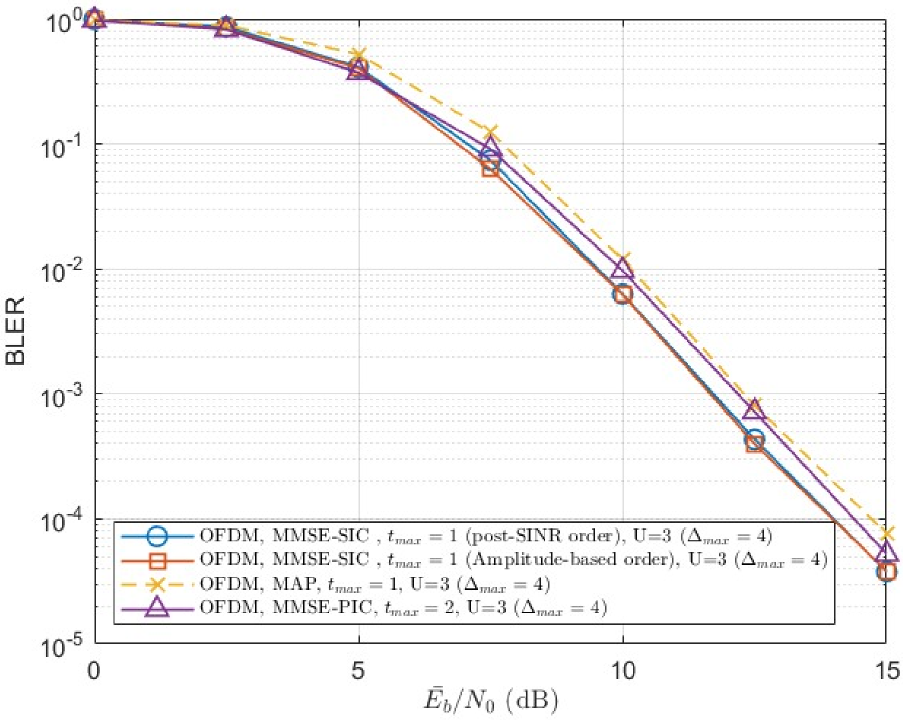

Figure 4.19: BLER of asynchronous OFDM system with MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD. $R = 2$, $U = 3$, $\Delta_{\max} = 4$, $t_{\max} = 1$ for MMSE-SIC-IDD and MAP-IDD.

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less robust to asynchronous effects.

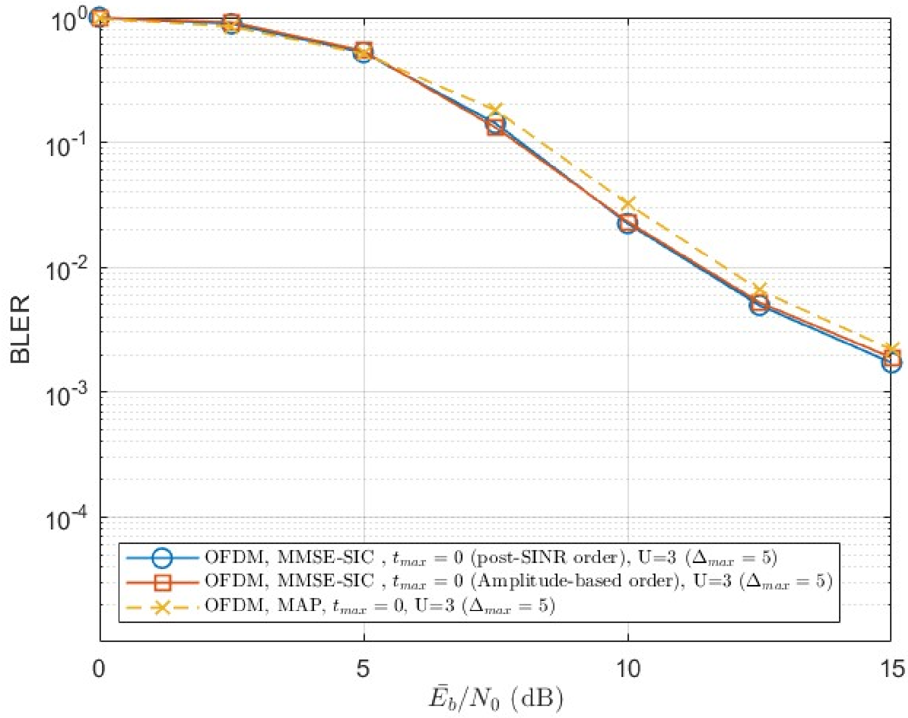

Figure 4.20: BLER of asynchronous OFDM system with MMSE-SIC-IDD and MAP-IDD. $R = 2$, $U = 3$, $\Delta_{\max} = 5$, $t_{\max} = 0$.

Figures 4.22 and 4.23 present the four-user case with $\Delta_{\max} = 4$. When $t_{\max} = 0$, MAP-IDD achieves the best high-SNR BLER performance because heavier loading leads to more severe residual interference. This is consistent with the synchronous OFDM results, where MAP-IDD is more stable under heavier loading. In Figure 4.23, MMSE-PIC-IDD with $t_{\max} = 2$ becomes slightly worse than MAP-IDD with $t_{\max} = 1$. This differs from the three-user case, where MMSE-PIC-IDD slightly outperforms MAP-IDD. In contrast, MMSE-SIC-IDD with dynamic post-SINR ordering achieves BLER performance slightly better than MAP-IDD, which is opposite to the synchronous case. This suggests the robustness of MMSE-SIC-IDD with a priori information update to asynchronous effects even in the four-user case.

The case of $\Delta_{\max} = 5$ in Figures 4.24 and 4.25 follows a similar trend with $\Delta_{\max} = 4$

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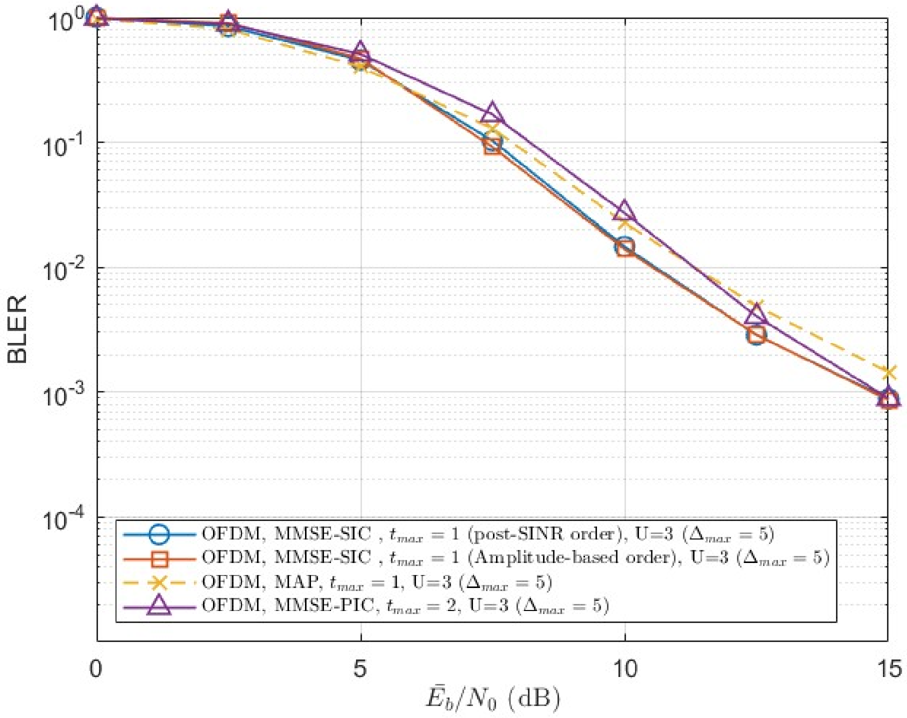

Figure 4.21: BLER of asynchronous OFDM system with MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD. $R = 2$, $U = 3$, $\Delta_{\max} = 5$, $t_{\max} = 1$ for MMSE-SIC-IDD and MAP-IDD.

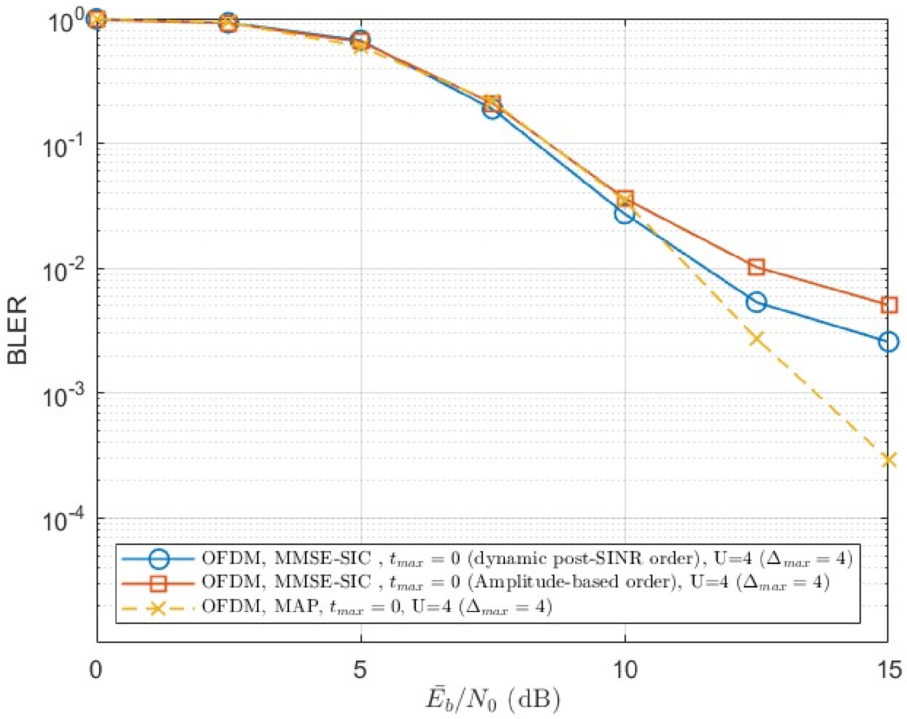

Figure 4.22: BLER of asynchronous OFDM system with MMSE-SIC-IDD and MAP-IDD. $R = 2$, $U = 4$, $\Delta_{\max} = 4$, $t_{\max} = 0$.

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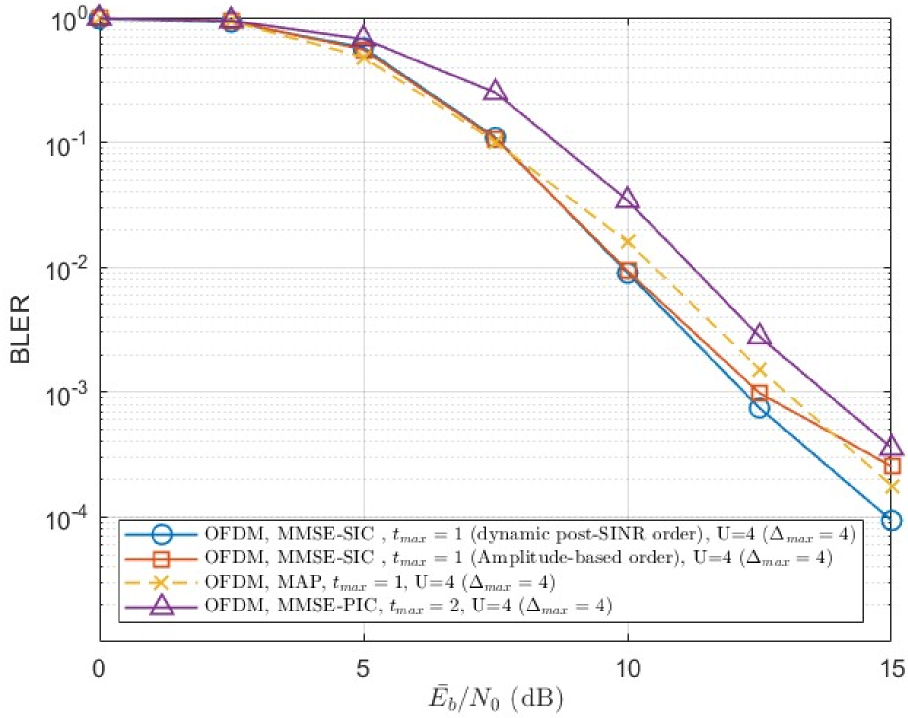

Figure 4.23: BLER of asynchronous OFDM system with MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD. $R=2$, $U=4$, $\Delta_{\max}=4$, $t_{\max}=1$ for MMSE-SIC-IDD and MAP-IDD.

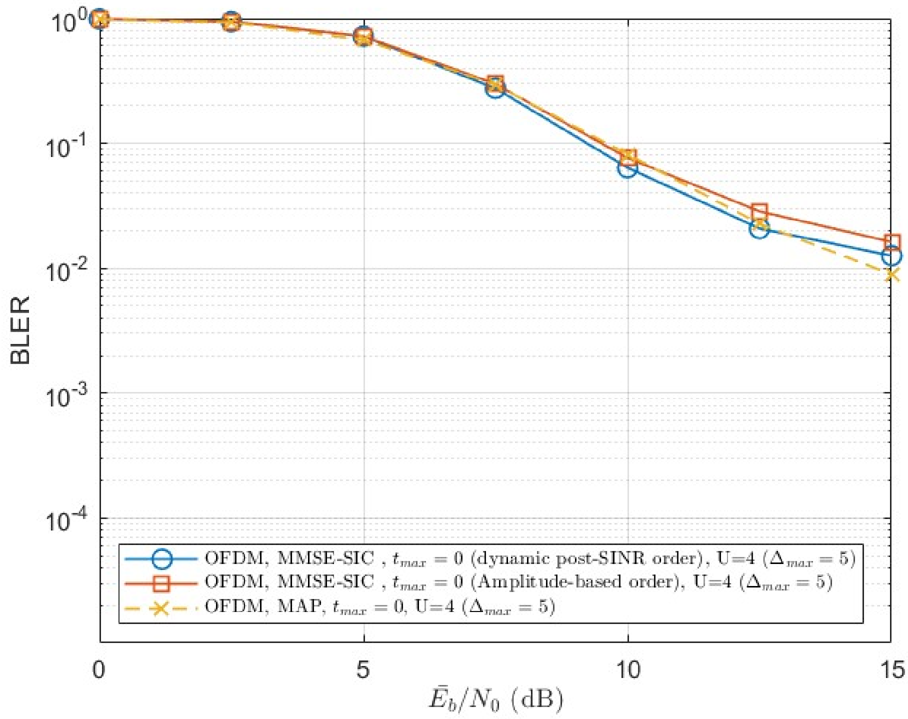

Figure 4.24: BLER of asynchronous OFDM system with MMSE-SIC-IDD and MAP-IDD. $R=2$, $U=4$, $\Delta_{\max}=5$, $t_{\max}=0$.

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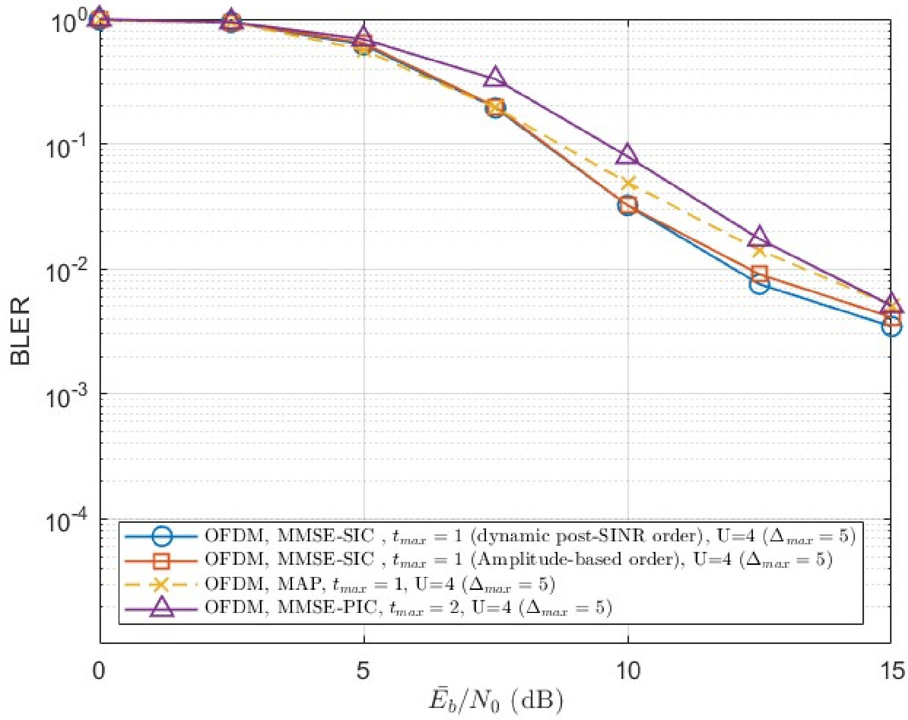

Figure 4.25: BLER of asynchronous OFDM system with MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD. $R = 2$, $U = 4$, $\Delta_{\text{max}} = 5$, $t_{\text{max}} = 1$ for MMSE-SIC-IDD and MAP-IDD.

while the overall BLER becomes worse. Under this condition, the performance is dominated not only by multiuser detection capability but also by the receiver's tolerance to residual MAI/ISI effects.

4.8 Comparison and summary

This chapter compares MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD in coded multi-resource NOMA and OFDM NOMA systems. In the coded multi-resource system, MAP-IDD provides the strongest detection capability, while MMSE-SIC-IDD offers a lower-complexity detection structure alternative through ordered linear filtering, decoder-aided cancellation, and variance-aware soft information. MMSE-PIC-IDD is more parallel, but its performance strongly depends on reliable a priori information from the decoder.

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In the synchronous OFDM system, subcarrier-dependent channel diversity improves the user separation capability of MMSE-SIC-IDD. With post-SINR ordering and one a priori information update, MMSE-SIC-IDD can slightly outperform MAP-IDD in the three-user case. However, as the number of users increases, MAP-IDD remains more stable, whereas MMSE-SIC-IDD becomes more sensitive to error propagation.

In the asynchronous OFDM system, the advantage of MAP-IDD becomes less pronounced because the assumed phase-rotated frequency-domain model may not fully match the actual received signal. MMSE-SIC-IDD shows stronger robustness in several asynchronous cases because the undecoded users are modeled as interference-plus-noise, allowing part of the asynchronous mismatch to be absorbed into the effective interference term. Overall, MMSE-SIC-IDD provides a favorable complexity-performance tradeoff, MMSE-PIC-IDD is useful when reliable a priori information is available, and MAP-IDD remains a high-performance receiver, especially under heavier loading and perfect synchronous models.

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Chapter 5 Performance analysis of user pairing based on CQI

The previous chapters evaluated the reliability of uplink NOMA receivers from the link-level BER or BLER viewpoint. This chapter further studies a practical system-level resource-sharing problem, where two OMA users are paired to share one resource.

By pairing two users on the same resource and adopting multi-user detection (MUD), one resource can be released for an additional user. However, the paired users need to reduce their modulation and coding scheme (MCS) because multi-user interference exists. Therefore, the objective of this chapter is not to show that the paired users always obtain a larger sum spectral efficiency (SE). Instead, we evaluate the tradeoff between the MCS degradation of the paired users and the additional SE brought by the released resource.

The analysis is conducted from an MCS-level viewpoint. First, a channel quality indicator (CQI)-like MCS table is constructed through link-level simulations. Then, two-user MUD tables are obtained for GDMA and soft SIC receivers. Based on these tables, an offline CQI-based pairing table is built to determine whether two OMA users can share one resource while maintaining the target BLER.

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5.1 CQI-like MCS table

CQI is commonly used as a channel-quality feedback indicator for link adaptation [29]. A higher CQI corresponds to a better channel condition and allows the system to select a higher MCS. In this work, we follow this CQI-based link-adaptation concept and construct a CQI-like table from the considered link-level simulation under the specified conditions. This is necessary because the channel model, channel code, and MUD receiver we considered differ from those of a standardized link-adaptation table [30].

The single-user received signal is modeled as

$$y_i = \sqrt{\beta_i} h_i x_i + n, \tag{5.1}$$

where β_i denotes the large-scale coefficient associated with CQI i, h_i is the fading coefficient, x_i is the transmitted symbol and $n \sim \mathcal{CN}(0, N_0)$ is complex AWGN. For each MCS, the operating E_s/N_0 is swept under a single-carrier block Rayleigh fading channel with an LDPC code of length $N = 1296$. The considered code rates are $1/2$, $2/3$, and $3/4$. The SNR threshold $\gamma_{\text{th},i}$ is chosen at which the BLER reaches 10^{-1}. The corresponding single-user received-SNR threshold is

$$\gamma_{\text{th},i} = \frac{\beta_i E_s}{N_0}. \tag{5.2}$$

The spectral efficiency of each MCS is defined as the number of coded bits per modulation symbol multiplied by the code rate.

The CQI table is normalized by using CQI 1 as the reference operating point. Specifically, CQI 1 is assigned $\beta_1 = 1$ and a coverage radius of $R_1 = 5$ km. With the simulated

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Table 5.1: CQI-like MCS table obtained from single-user link-level simulation. CQI 3 is listed for completeness but is not used in the proposed scheme.

CQI	MCS	SE (bit/s/Hz)	γ_{th} (dB)	β	R (km)
1	BPSK, $1/2$	0.50	7.50	1.0000	5.0000
2	BPSK, $2/3$	0.67	9.30	1.5136	4.0641
~~3~~	~~BPSK, $3/4$~~	~~0.75~~	~~10.76~~	~~2.1184~~	~~3.4353~~
4	QPSK, $1/2$	1.00	10.46	1.9770	3.5560
5	QPSK, $2/3$	1.33	12.56	3.2063	2.7923
6	QPSK, $3/4$	1.50	13.741	4.2082	2.4374
7	16QAM, $1/2$	2.00	15.679	6.5751	1.9499
8	16QAM, $2/3$	2.67	18.418	12.3540	1.4225
9	16QAM, $3/4$	3.00	19.703	16.6070	1.2269

threshold of CQI 1, the noise variance is obtained from (5.2) as

$$N_0 = \frac{\beta_1 E_s}{10^{\gamma_{\text{th},1}/10}}. \tag{5.3}$$

The large-scale coefficient of each CQI level is calculated by

$$\beta_i = \frac{N_0}{E_s} 10^{\gamma_{\text{th},i}/10}. \tag{5.4}$$

We adopt the path-loss model $\beta_i \propto 1/R_i^2$. Therefore, the corresponding radius of CQI i is

$$R_i = \frac{R_1}{\sqrt{\beta_i}}. \tag{5.5}$$

Table 5.1 lists the resulting CQI-like MCS table. CQI 3 is dominated by CQI 4 because it provides a lower SE but requires a higher SNR threshold. Therefore, the remaining valid CQI set is

$$\mathcal{Q} = \{1, 2, 4, 5, 6, 7, 8, 9\}. \tag{5.6}$$

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5.2 Two-user MUD pairs scheme

We next construct the two-user MUD pairing-state table. The purpose of this table is to determine, for each possible after-pair MCS combination, the aggregate SNR required by the receiver to make all users satisfy the BLER constraint.

For two users sharing the same resource, the received signal is written as

$$y = \sqrt{\beta_i} h_1 x_1 + \sqrt{\beta_j} h_2 x_2 + n. \tag{5.7}$$

When two OMA users are paired on one resource and detected by a MUD receiver, the receiver can use the received signal energies of both users. Hence, the effective SNR available for the paired transmission is defined as the aggregate received SNR

$$\Gamma_{i,j}^{\mathrm{agg}} = \frac{(\beta_i + \beta_j) E_s}{N_0} = \frac{\sum_{u=1}^{2} \beta_u E_s}{N_0}. \tag{5.8}$$

For each candidate after-pair CQI state (p, q), GDMA detection and soft SIC detection are simulated by setting the noise power according to (5.8). The required threshold $\Gamma_{\mathrm{th},p,q}^{\mathrm{agg}}$ is the minimum aggregate SNR such that all users satisfy

$$\mathrm{BLER}_u \leq 10^{-1}, \qquad u = 1, 2. \tag{5.9}$$

After simulating the candidate MCS combinations, dominated combinations are removed. A candidate is considered dominated if another candidate provides a higher or equal sum SE with a lower required aggregate-SNR threshold. The remaining candidates form the effective MUD pairing-state table. For GDMA, the complete candidate table before removing dominated combinations is listed in Appendix A.

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Table 5.2: Effective GDMA after-pairing states after removing dominated MCS combinations.

After-pair CQI state	Sum SE (bit/s/Hz)	Required $\Gamma_{\text{th}}^{\text{agg}}$ (dB)
$(1, 1)$	1.00	11.88
$(1, 2)$	1.17	14.02
$(2, 2)$	1.34	14.52
$(1, 4)$	1.50	14.58
$(1, 5)$	1.83	15.53
$(4, 4)$	2.00	16.70
$(2, 6)$	2.17	18.27
$(4, 5)$	2.33	19.56
$(5, 5)$	2.66	20.02
$(5, 6)$	2.83	21.22
$(6, 6)$	3.00	21.62

Table 5.3: Effective soft SIC after-pairing states.

After-pair CQI state	Sum SE (bit/s/Hz)	Required $\Gamma_{\text{th}}^{\text{agg}}$ (dB)
$(1, 1)$	1.00	12.39
$(1, 2)$	1.17	14.83
$(2, 2)$	1.34	16.967
$(1, 4)$	1.50	17.015
$(2, 4)$	1.67	18.19

Tables 5.2 and 5.3 show that GDMA supports more effective after-pairing states and requires less SNR $\Gamma_{\text{th}}^{\text{agg}}$ compared with soft SIC. This difference directly affects the user-pairing performance, because a receiver with a stronger MUD capability can keep a higher after-pair MCS and reduce the SE loss caused by pairing.

5.3 Offline CQI-based pairing table

The offline pairing table maps two original OMA CQI indices to a feasible after-pair CQI state. Suppose two users originally use two orthogonal resources with CQI indices i and j. The OMA before-pairing sum SE is

$$\eta_{i,j}^{\text{OMA}} = \eta_i + \eta_j, \tag{5.10}$$

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where η_i is the SE of CQI i in Table 5.1. If the users are paired on one shared resource, the aggregate received SNR $\Gamma_{i,j}^{\text{agg}}$ in (5.8) is computed from their received-power coefficients. The after-pair CQI state is then selected as the highest-SE state in the corresponding MUD pairing-state table whose required SNR is not larger than the available aggregate SNR.

Let (p, q) denote the selected after-pair CQI state. The after-pair sum SE is

$$\eta_{p,q}^{\text{NOMA}} = \eta_p + \eta_q. \tag{5.11}$$

The paired users' SE loss is defined as the positive quantity

$$D_{i,j} = \eta_{i,j}^{\text{OMA}} - \eta_{p,q}^{\text{NOMA}} \geq 0. \tag{5.12}$$

The system-level benefit of pairing is then determined by whether the SE of the newly admitted user can compensate for $D_{i,j}$.

The complete offline pairing tables for GDMA and soft SIC are listed in Tables 5.4 and 5.5, respectively. For example, if two OMA users have CQI 2 and CQI 9, their OMA sum SE is $0.67 + 3 = 3.67$ bit/s/Hz over two resources. Under GDMA detection, their aggregate received SNR is sufficient to select the after-pair state $(5, 5)$, whose sum SE is 2.66 bit/s/Hz over one resource. Thus, the paired users lose $D_{2,9} = 1.01$ bit/s/Hz, but one resource is released for another user. Under soft SIC, the same pair is mapped to $(2, 4)$, and the loss becomes $D_{2,9} = 2.00$ bit/s/Hz. This example illustrates why GDMA is more favorable for CQI-based pairing.

Although the paired users' sum SE may decrease after pairing, the SE per occupied resource can still increase because two OMA resources are compressed into one shared resource. Therefore, we also evaluate the SE increment per resource. For a before-pair

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Table 5.4: Offline pairing table for GDMA detection.

Before CQI pair	Before SE	SNR (dB)	After CQI state	$D_{i,j}$
$(1,4)$	1.50	12.2377	$(1,1)$	0.50
$(1,5)$	1.83	13.7390	$(1,1)$	0.83
$(1,6)$	2.00	14.6668	$(1,4)$	0.50
$(1,7)$	2.50	16.2938	$(1,5)$	0.67
$(1,8)$	3.17	18.7561	$(2,6)$	1.00
$(1,9)$	3.50	19.9568	$(4,5)$	1.17
$(2,2)$	1.34	12.3104	$(1,1)$	0.34
$(2,4)$	1.67	12.9290	$(1,1)$	0.67
$(2,5)$	2.00	14.2393	$(1,2)$	0.83
$(2,6)$	2.17	15.0753	$(1,4)$	0.67
$(2,7)$	2.67	16.5787	$(1,5)$	0.84
$(2,8)$	3.34	18.9200	$(2,6)$	1.17
$(2,9)$	3.67	20.0817	$(5,5)$	1.01
$(4,4)$	2.00	13.4703	$(1,1)$	1.00
$(4,5)$	2.33	14.6460	$(1,4)$	0.83
$(4,6)$	2.50	15.4135	$(1,4)$	1.00
$(4,7)$	3.00	16.8207	$(4,4)$	1.00
$(4,8)$	3.67	19.0627	$(2,6)$	1.50
$(4,9)$	4.00	20.1913	$(5,5)$	1.34
$(5,5)$	2.66	15.5700	$(1,5)$	0.83
$(5,6)$	2.83	16.2008	$(1,5)$	1.00
$(5,7)$	3.33	17.4040	$(4,4)$	1.33
$(5,8)$	4.00	19.4201	$(2,6)$	1.83
$(5,9)$	4.33	20.4695	$(5,5)$	1.67
$(6,6)$	3.00	16.7512	$(4,4)$	1.00
$(6,7)$	3.50	17.8275	$(4,4)$	1.50
$(6,8)$	4.17	19.6911	$(4,5)$	1.84
$(6,9)$	4.50	20.6838	$(5,5)$	1.84
$(7,7)$	4.00	18.6893	$(2,6)$	1.83
$(7,8)$	4.67	20.2712	$(5,5)$	2.01
$(7,9)$	5.00	21.1515	$(5,5)$	2.34
$(8,8)$	5.34	21.4283	$(5,6)$	2.51
$(8,9)$	5.67	22.1181	$(6,6)$	2.67
$(9,9)$	6.00	22.7132	$(6,6)$	3.00

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Table 5.5: Offline pairing table for soft SIC detection.

Before CQI pair	Before SE	SNR (dB)	After CQI state	$D_{i,j}$
$(1,5)$	1.83	13.7390	$(1,1)$	0.83
$(1,6)$	2.00	14.6668	$(1,1)$	1.00
$(1,7)$	2.50	16.2938	$(1,2)$	1.33
$(1,8)$	3.17	18.7561	$(2,4)$	1.50
$(1,9)$	3.50	19.9568	$(2,4)$	1.83
$(2,4)$	1.67	12.9290	$(1,1)$	0.67
$(2,5)$	2.00	14.2393	$(1,1)$	1.00
$(2,6)$	2.17	15.0753	$(1,2)$	1.00
$(2,7)$	2.67	16.5787	$(1,2)$	1.50
$(2,8)$	3.34	18.9200	$(2,4)$	1.67
$(2,9)$	3.67	20.0817	$(2,4)$	2.00
$(4,4)$	2.00	13.4703	$(1,1)$	1.00
$(4,5)$	2.33	14.6460	$(1,1)$	1.33
$(4,6)$	2.50	15.4135	$(1,2)$	1.33
$(4,7)$	3.00	16.8207	$(1,2)$	1.83
$(4,8)$	3.67	19.0627	$(2,4)$	2.00
$(4,9)$	4.00	20.1913	$(2,4)$	2.33
$(5,5)$	2.66	15.5700	$(1,2)$	1.49
$(5,6)$	2.83	16.2008	$(1,2)$	1.66
$(5,7)$	3.33	17.4040	$(1,4)$	1.83
$(5,8)$	4.00	19.4201	$(2,4)$	2.33
$(5,9)$	4.33	20.4695	$(2,4)$	2.66
$(6,6)$	3.00	16.7512	$(1,2)$	1.83
$(6,7)$	3.50	17.8275	$(1,4)$	2.00
$(6,8)$	4.17	19.6911	$(2,4)$	2.50
$(6,9)$	4.50	20.6838	$(2,4)$	2.83
$(7,7)$	4.00	18.6893	$(2,4)$	2.33
$(7,8)$	4.67	20.2712	$(2,4)$	3.00
$(7,9)$	5.00	21.1515	$(2,4)$	3.33
$(8,8)$	5.34	21.4283	$(2,4)$	3.67
$(8,9)$	5.67	22.1181	$(2,4)$	4.00
$(9,9)$	6.00	22.7132	$(2,4)$	4.33

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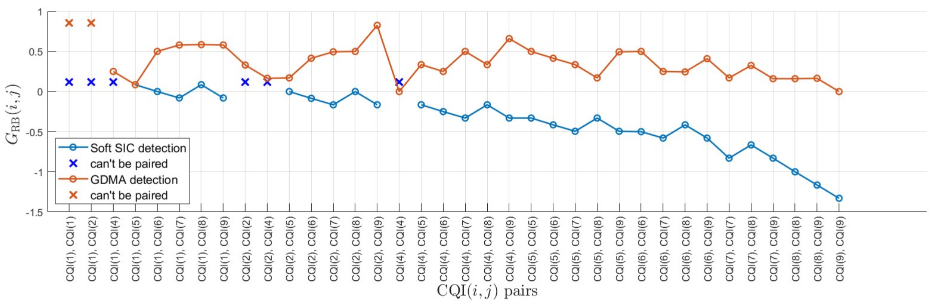

Figure 5.1: SE increment per resource for GDMA and soft SIC.

CQI pair (i, j) mapped to the after-pair CQI state (p, q), the OMA average SE per resource is $(\eta_i + \eta_j)/2$, while the shared-resource SE after pairing is $\eta_p + \eta_q$. The per-resource SE increment is defined as

$$G_{\mathrm{RB}}(i,j) = \eta_p + \eta_q - \frac{\eta_i + \eta_j}{2} \tag{5.13}$$

The comparison for GDMA and soft SIC will be shown in Fig. 5.1. This result shows that GDMA can provide a larger per-resource SE increment than soft SIC in most of CQI pairing cases.

5.4 System-level pairing operation

The system-level simulation starts from $N_0^{\text{user}} = 100$ OMA users. Each initial user occupies one resource. Whenever two users are paired on a single resource, one resource is released and can be assigned to a newly admitted user. Figure 5.2 illustrates this operation.

Let S_{100} denote the total SE of the initial 100 OMA users. If P pairs are formed, each pairing event releases one resource and creates one opportunity for a new user. In Case 1, the CQI of a new user is unknown before admission and is drawn randomly from the considered CQI distribution. Let A be the average SE of a new user. The SE increment

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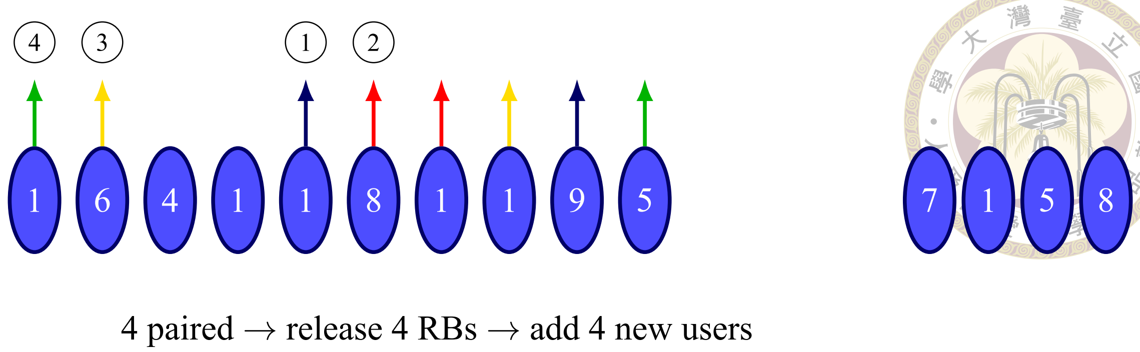

Figure 5.2: Illustration of the system operation. Assume that there are initially 10 users with their corresponding CQI values. The pairing order follows the four colored pairings labeled 1–4; each pairing releases one resource. After admitting four new users, the remaining users, together with the newly admitted users, continue to be paired and release resources until no further pairing is possible.

after P pairings is defined as

$$G_{\text{SE}}^{(1)}(P) = \frac{\sum_{\xi=1}^{P}\left(A - D_{i_\xi, j_\xi}\right)}{S_{100}} \times 100\%, \qquad P < 100. \tag{5.14}$$

In Case 2, the system has prior knowledge of the CQI values of the users waiting for admission and the number of waiting users is N_{new}. If the ξth admitted new user has SE $\eta_{\text{new},\xi}$, the SE increment is

$$G_{\text{SE}}^{(2)}(P) = \frac{\sum_{\xi=1}^{P}\left(\eta_{\text{new},\xi} - D_{i_\xi, j_\xi}\right)}{S_{100}} \times 100\%, \qquad P < N_{\text{new}}. \tag{5.15}$$

The value of G_{SE} can be negative if the paired users' MCS degradation is larger than the SE contributed by the newly admitted users.

5.4.1 Pairing rules

Three pairing rules are considered [31].

1. **Rule 1: maximizing the number of served users.** This rule pairs as many users as possible according to the offline pairing table. Users with high CQI values are

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prioritized to pair with users having low CQI values. The rule is user-count oriented and does not explicitly constrain the SE loss.

2. **Rule 2: maximizing system SE with a threshold.** This rule forms a pair only when the paired users' SE loss satisfies

$$D_{i,j} \leq T, \tag{5.16}$$

where T is the predefined loss threshold. In the simulation, T can be chosen according to the expected SE of a newly admitted user A, so that only pairings with non-negative expected SE contribution are preferred.

3. **Rule 3: maximizing system SE with known new-user CQI.** This rule is used in Case 2. Since the CQI of each waiting user is known, a pair is formed only when

$$D_{i,j} \leq \eta_{\text{new}}. \tag{5.17}$$

Therefore, the released resource is used only when the admitted new user can compensate for the paired users' SE loss.

The simulation results are averaged over 1000 independent runs. The CQI distributions follow the uniform and nonuniform user-density [31]. Two distributions are listed in Table 5.6. The uniform distributed case assumes that the user density is proportional to $1/R^2$, which results in more users located in low-CQI regions. Therefore, this case is also described as the low-CQI-dominant distribution. The nonuniform distributed case assumes that the user density is proportional to $1/R$, which increases the relative probability of users in high-CQI regions. Therefore, this case is also described as the high-CQI-dominant distribution.

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Table 5.6: CQI distributions used in the system-level simulation.

CQI	SE	Uniform distributed (probability)	Nonuniform distributed (probability)
1	0.50	0.33932	0.056713
2	0.67	0.15486	0.069774
4	1.00	0.19393	0.079742
5	1.33	0.074255	0.10155
6	1.50	0.085542	0.11634
7	2.00	0.071143	0.14542
8	2.67	0.020730	0.19934
9	3.00	0.060216	0.23112
Average		$A = 1.0727$	$A = 1.9809$

5.5 Simulation results

5.5.1 Case 1: no upper limit on the number of users

In Case 1, there is no upper limit on the number of users in the system. Whenever a resource is released, a new user is admitted randomly according to the corresponding CQI distribution. Therefore, the value of a released resource is represented by the average new-user SE A.

Figures 5.3 and 5.4 compare GDMA and soft SIC in Case 1 with the low-CQI-dominant and high-CQI-dominant distributions, respectively. Under Rule 1, both receivers can admit the full number of users because the rule attempts to pair users whenever a feasible entry exists in the offline pairing table. However, the SE behavior is receiver dependent. GDMA maintains a positive SE increment because its after-pair MCS states are stronger and the resulting loss $D_{i,j}$ is smaller. Soft SIC can admit users, but many pairings cause a large SE loss and the overall SE increment can become negative, especially when high-CQI users are more frequent.

Rule 2 improves the SE increment by rejecting pairings whose loss exceeds the

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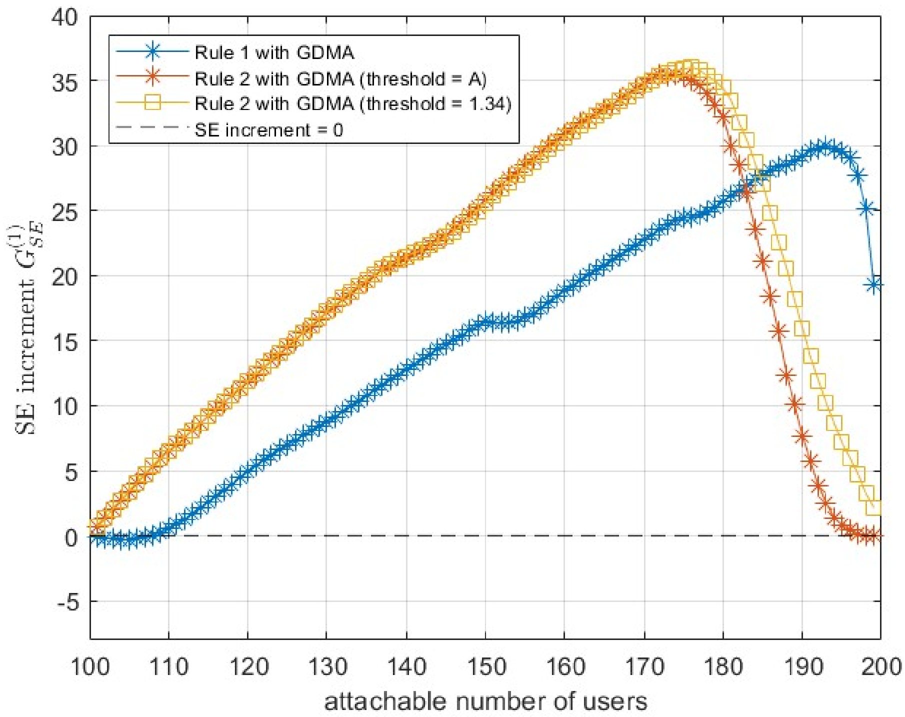
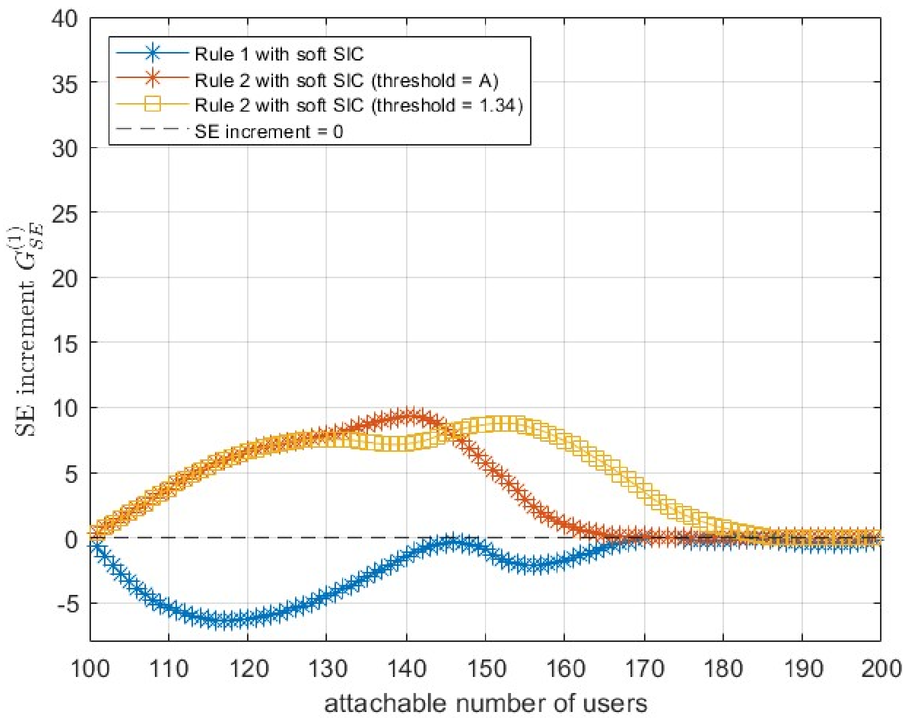


(a) GDMA detection



(b) Soft SIC detection

Figure 5.3: Case 1 with the low-CQI-dominant distribution.

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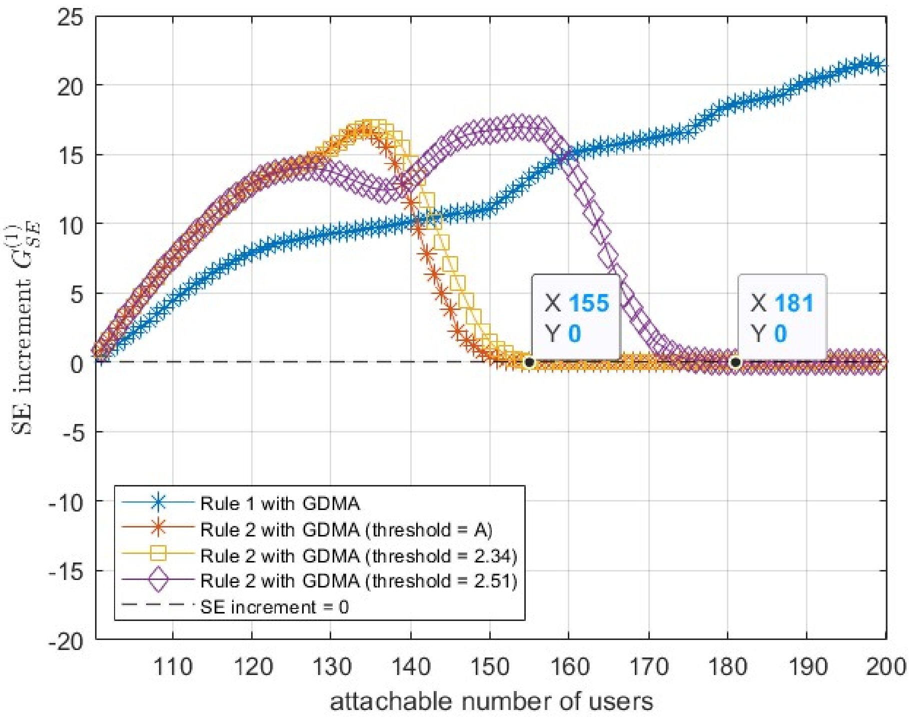
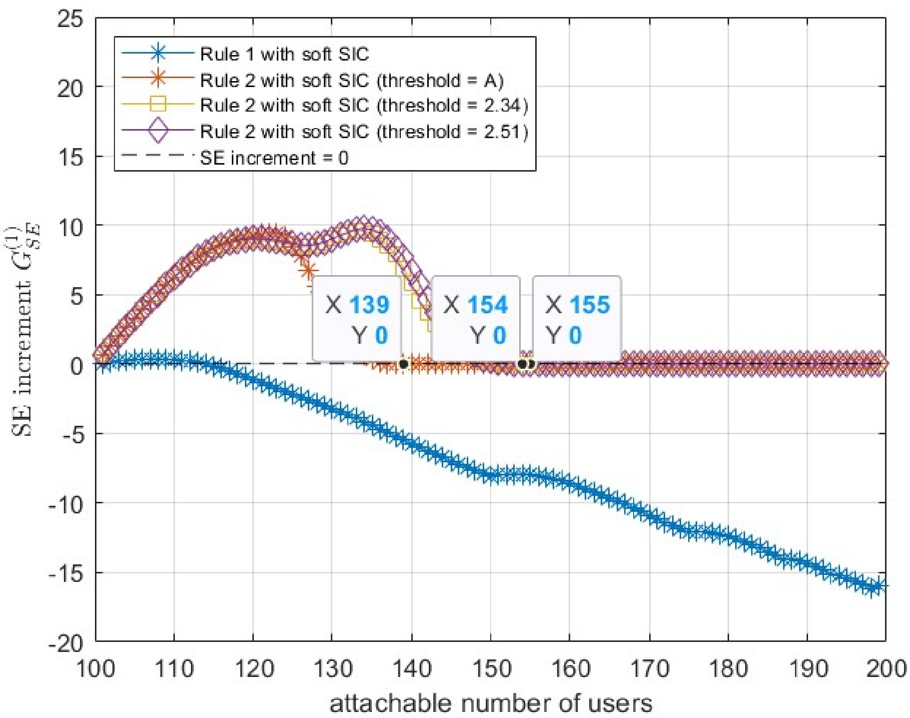


(a) GDMA detection



(b) Soft SIC detection

Figure 5.4: Case 1 with the high-CQI-dominant distribution.

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threshold, leading to a gain-then-collapse behavior. In the moderate-loading region, Rule 2 can yield a higher SE increment than Rule 1 because inefficient pairings are suppressed. Once the remaining feasible pairs all violate the threshold, the scheme stops forming new pairs and the SE increment drops to zero. The cutoff occurs earlier under the high-CQI-dominant distribution, since pairing high-CQI users tends to cause a larger MCS degradation.

It can also be observed that, as the threshold T increases, the SE increment improves slightly and the cutoff occurs later. Although a larger threshold permits more pairings and thus extends the feasible region, it also allows more pairings, which can reduce the SE increment in the moderate-loading region.

5.5.2 Case 2: fixed maximum number of new users

In Case 2, the maximum number of new users is fixed as $N_{\text{new}} = 100$. The system is assumed to know the CQI values of the users waiting for admission. Hence, high-CQI new users can be admitted first, and Rule 3 can compare the actual SE of the new user with the loss of the candidate pair.

Figures 5.5 and 5.6 show the Case 2 results. Since high-CQI new users are admitted first, the released resources have a high immediate value. Rule 3 therefore provides a large SE gain in the moderate-loading region. For the low-CQI-dominant distribution of 5.5, soft SIC reaches about 33% peak SE increment, while GDMA reaches about 45%. After the feasible pairing set is exhausted, the SE increment quickly drops to zero. The cutoff occurs at approximately 171 served users for soft SIC and 179 served users for GDMA.

The gap between GDMA and soft SIC is smaller under Rule 3 than under Rule 1.

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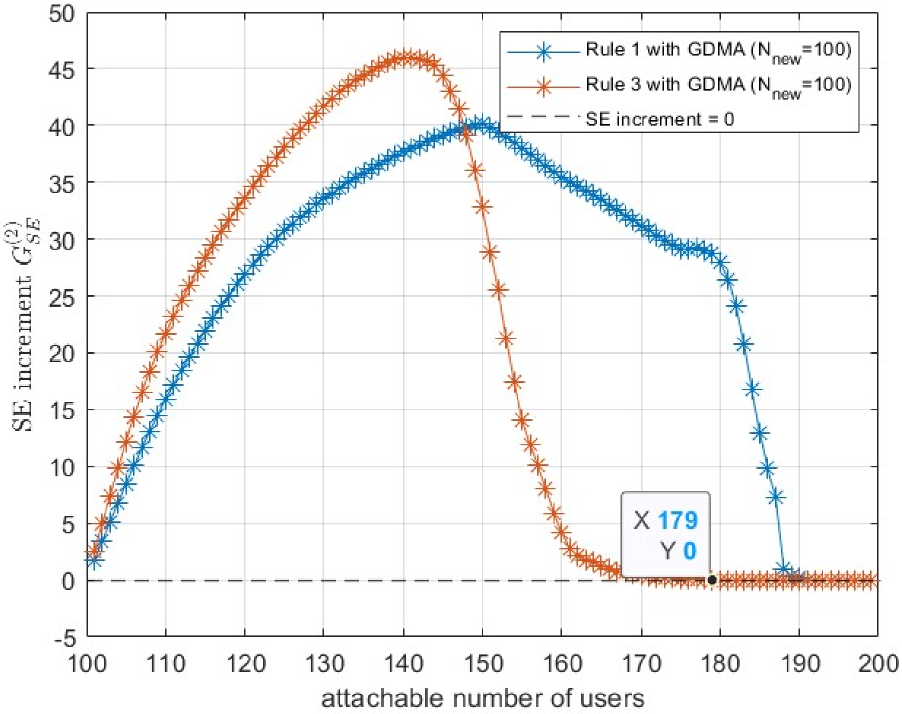
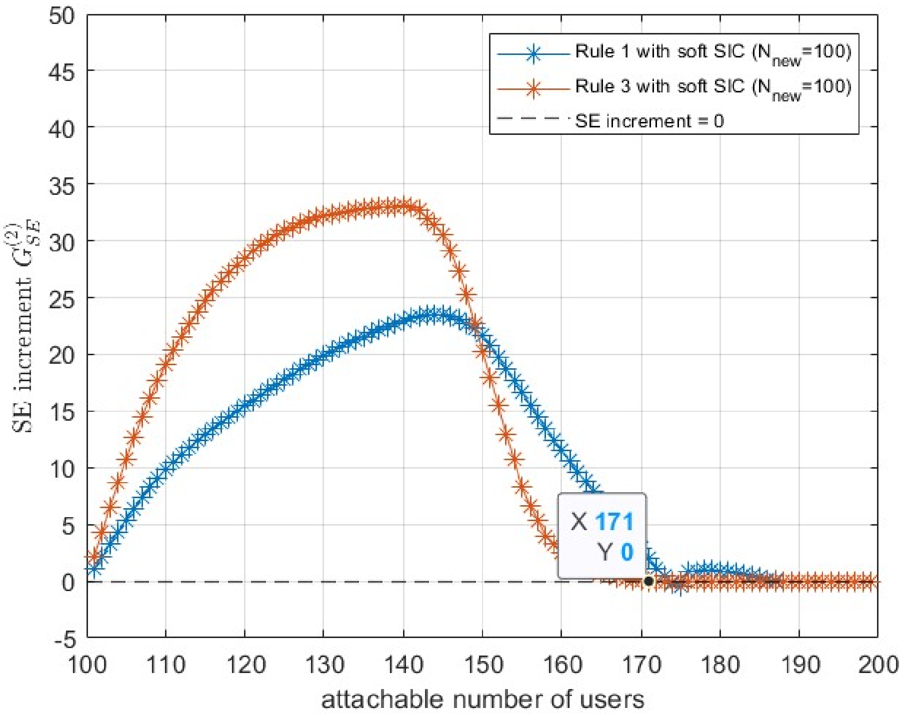


(a) GDMA detection



(b) Soft SIC detection

Figure 5.5: Case 2 with the low-CQI-dominant distribution.

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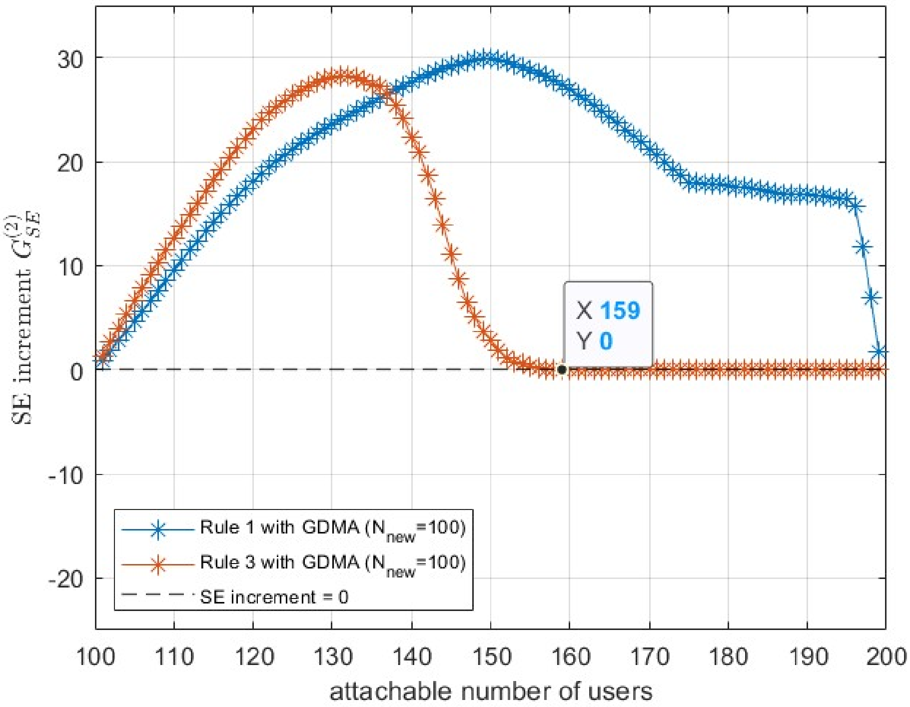
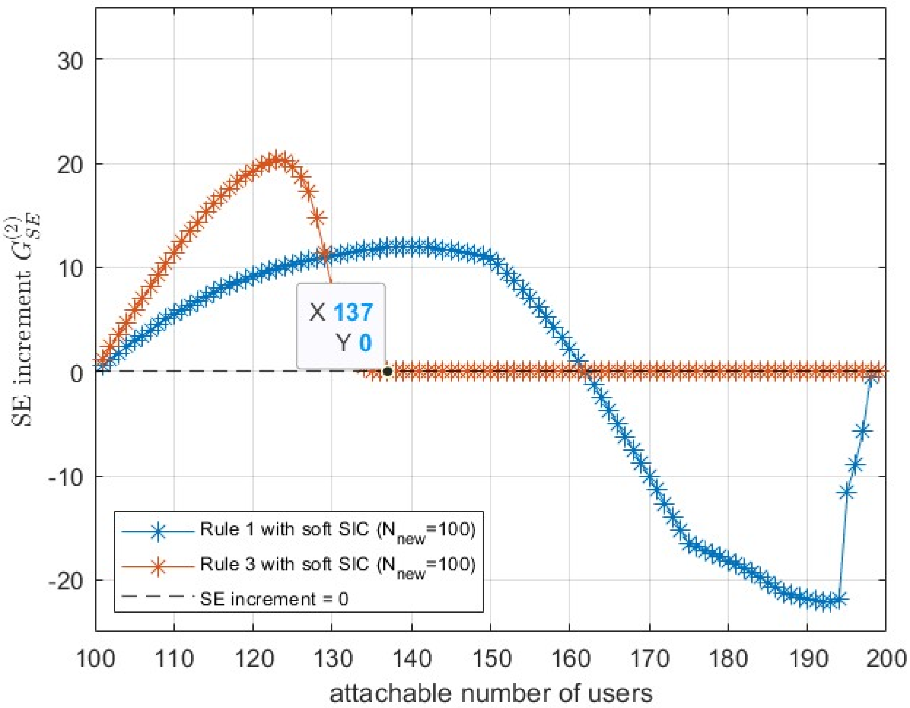


(a) GDMA detection



(b) Soft SIC detection

Figure 5.6: Case 2 with the high-CQI-dominant distribution.

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The reason is that Rule 3 admits high-CQI new users first and only accepts pairings with sufficiently small SE loss. For such low-loss pairs, the selected after-pair CQI states of the two receivers are often close. Nevertheless, under heavy loading or under Rule 1, GDMA remains more robust because its offline pairing table contains more effective after-pairing states and smaller SE losses.

5.6 Summary

This chapter develops a practical CQI-based user-pairing analysis for NOMA over OMA. The main idea is to evaluate NOMA pairing at the MCS level instead of relying only on ideal capacity formulas. A CQI-like MCS table is first built from link-level BLER simulation. Then, GDMA and soft SIC MUD tables are constructed using the aggregate received SNR definition in (5.8). These MUD tables are used to build offline pairing tables that map OMA before-pair CQI indices to NOMA after-pair CQI states.

The results show that NOMA pairing does not necessarily increase the sum SE of the paired users. The paired users usually suffer an MCS degradation. The practical gain comes from whether the released resource can be reused by a new user whose SE compensates for this loss. GDMA is more favorable than soft SIC because it supports more effective after-pairing states and causes smaller SE losses. Rule 1 maximizes the number of served users but can be SE-unfavorable for weak receivers. Rule 2 and Rule 3 improve the SE increment by suppressing pairings with excessive loss, but they also reduce the number of feasible pairings and can stop being applicable after the efficient pairing set is exhausted.

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Chapter 6 Conclusions and future work

In chapter 2,

In chapter 3,

In chapter 4,

In chapter 5,

In future research,

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Appendix A — Complete GDMA MUD Candidate Table

A.1 Complete GDMA Candidate MCS Combinations

Table A.1 lists the complete GDMA candidate MCS combinations simulated under the aggregate received-SNR definition in (5.8). Each entry gives the after-pair CQI state, the corresponding sum SE, and the required aggregate SNR such that all users satisfy BLER $\leq 10^{-1}$. In Chapter 5, the dominated combinations are removed from this table to obtain the effective GDMA after-pairing states in Table 5.2.

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Table A.1: Complete GDMA after-pairing candidate states before removing dominated MCS combinations.

After-pair CQI state	Sum SE (bit/s/Hz)	Required $\Gamma_{\text{th}}^{\text{agg}}$ (dB)
$(1,1)$	1.00	11.88
$(1,2)$	1.17	14.02
$(1,4)$	1.50	14.58
$(1,5)$	1.83	15.53
$(1,6)$	2.00	17.20
$(1,7)$	2.50	22.31
$(1,8)$	3.17	23.80
$(1,9)$	3.50	24.42
$(2,2)$	1.34	14.52
$(2,4)$	1.67	16.94
$(2,5)$	2.00	17.56
$(2,6)$	2.17	18.27
$(4,4)$	2.00	16.70
$(4,5)$	2.33	19.56
$(4,6)$	2.50	21.28
$(5,5)$	2.66	20.02
$(5,6)$	2.83	21.22
$(6,6)$	3.00	21.62

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Appendix B — Introduction

B.1 Introduction

B.2 Further Introduction

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